

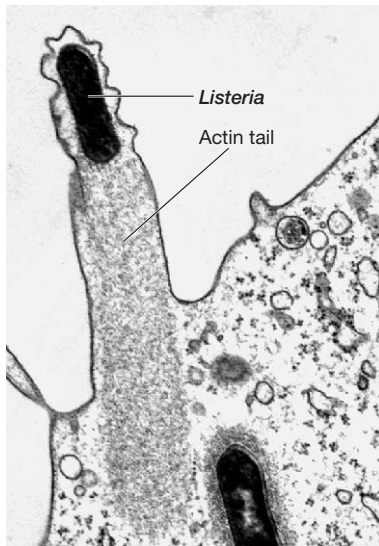


Figure 4.4 *Listeria* Motility and Actin Filaments. A *Listeria* cell is propelled through the cell surface by a bundle of actin filaments.

evidence. Cytochalasin B disrupts microfilament structure and often simultaneously inhibits cell movements. However, because the drug has additional effects in cells, a direct cause-and-effect interpretation of these experiments is sometimes difficult.

Microfilament protein has been isolated and analyzed chemically. It is an actin, very similar to the actin contractile protein of muscle tissue. This is further indirect evidence for microfilament involvement in cell movement.

Some pathogens such as *Listeria monocytogenes* make use of eucaryotic actin to move rapidly through the host cell. The ActA protein released by *Listeria* causes the polymerization of actin filaments at the end of the bacterium. A tail of actin is formed and trapped in the host cytoskeleton. Its continued elongation pushes the bacterium along at rates up to 11 $\mu\text{m}/\text{minute}$. The bacterium can even be propelled through the cell surface and into neighboring cells (**figure 4.4**).

A second type of small filamentous organelle in the cytoplasmic matrix is shaped like a thin cylinder about 25 nm in diameter. Because of its tubular nature this organelle is called a **microtubule**. Microtubules are complex structures constructed of two slightly different spherical protein subunits named tubulins, each of which is approximately 4 to 5 nm in diameter. These subunits are assembled in a helical arrangement to form a cylinder with an average of 13 subunits in one turn or circumference (**figure 4.5**).

Microtubules serve at least three purposes: (1) they help maintain cell shape, (2) are involved with microfilaments in cell movements, and (3) participate in intracellular transport processes. Evidence for a structural role comes from their intracellular distribution and studies on the effects of the drug colchicine. Long, thin cell structures requiring support such as the axopodia (long, slender, rigid pseudopodia) of protozoa contain microtubules (**figure 4.6**). When migrating embryonic nerve and

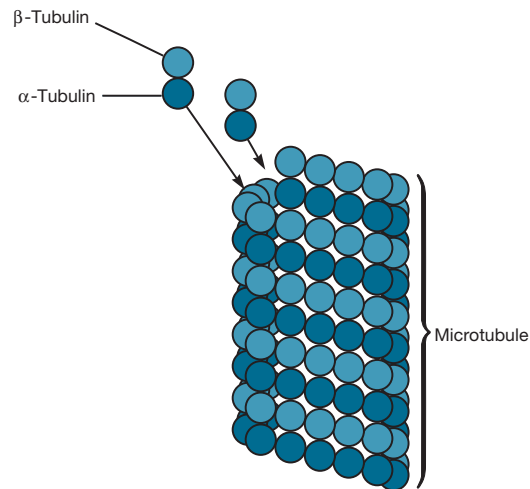
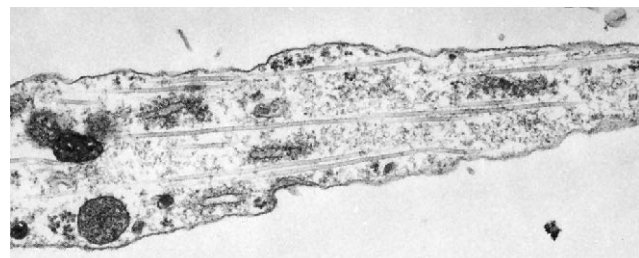
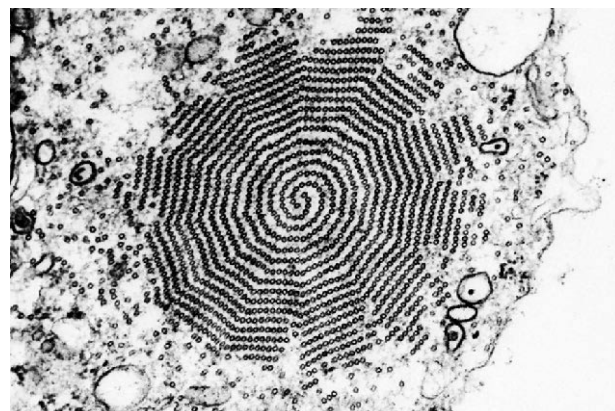


Figure 4.5 **Microtubule Structure.** The hollow cylinder, about 25 nm in diameter, is made of two kinds of protein subunits, α -tubulin and β -tubulin.



(a)



(b)

Figure 4.6 **Cytoplasmic Microtubules.** Electron micrographs of pseudopodia with microtubules. (a) Microtubules in a pseudopodium from the protozoan *Reticulomyxa* ($\times 65,000$). (b) A transverse section of a heliozoan axopodium ($\times 48,000$). Note the parallel array of microtubules organized in a spiral pattern.

heart cells are exposed to colchicine, they simultaneously lose their microtubules and their characteristic shapes. The shapeless cells seem to wander aimlessly as if incapable of directed movement without their normal form. Their microfilaments are still intact, but due to the disruption of their microtubules by colchicine, they no longer behave normally.

Microtubules also are present in structures that participate in cell or organelle movements—the mitotic spindle, cilia, and flagella. For example, the mitotic spindle is constructed of microtubules; when a dividing cell is treated with colchicine, the spindle is disrupted and chromosome separation blocked. Microtubules also are essential to the movement of eucaryotic cilia and flagella.

Other kinds of filamentous components also are present in the matrix, the most important of which are the **intermediate filaments** (about 8 to 10 nm in diameter). The microfilaments, microtubules, and intermediate filaments are major components of a vast, intricate network of interconnected filaments called the **cytoskeleton** (figure 4.7). As mentioned previously, the cytoskeleton plays a role in both cell shape and movement. Procaryotes lack a true, organized cytoskeleton and may not possess actinlike proteins.

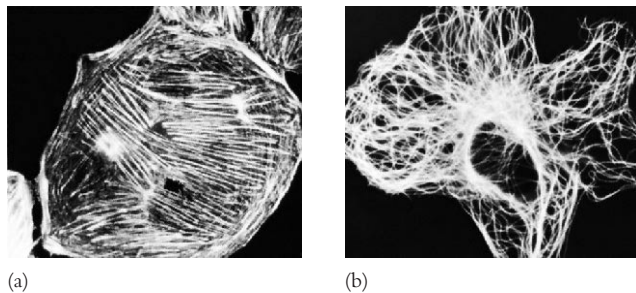


Figure 4.7 The Eucaryotic Cytoskeleton. (a) Antibody-stained microfilament system in a mammal cell ($\times 400$). (b) Antibody-stained microtubule system in a mammal cell ($\times 1,000$).

1. What is an organelle?
2. Define cytoplasmic matrix, bulk or free water, bound water, microfilament, microtubule, and tubulin. Discuss the roles of microfilaments, intermediate filaments, and microtubules.
3. Describe the cytoskeleton. What are its functions?

4.3 The Endoplasmic Reticulum

Besides the cytoskeleton, the cytoplasmic matrix is permeated with an irregular network of branching and fusing membranous tubules, around 40 to 70 nm in diameter, and many flattened sacs called **cisternae** (s., **cisterna**). This network of tubules and cisternae is the **endoplasmic reticulum (ER)** (figure 4.2a and figure 4.8). The nature of the ER varies with the functional and physiological status of the cell. In cells synthesizing a great deal of protein for purposes such as secretion, a large part of the ER is studded on its outer surface with ribosomes and is called **rough** or **granular endoplasmic reticulum (RER or GER)**. Other cells, such as those producing large quantities of lipids, have ER that lacks ribosomes. This is **smooth** or **agranular ER (SER or AER)**.

The endoplasmic reticulum has many important functions. It transports proteins, lipids, and probably other materials through the cell. Lipids and proteins are synthesized by ER-associated enzymes and ribosomes. Polypeptide chains synthesized on RER-bound ribosomes may be inserted either into the ER membrane or into its lumen for transport elsewhere. The ER is also a major site of cell membrane synthesis.

New endoplasmic reticulum is produced through expansion of the old. Many biologists think the RER synthesizes new ER proteins and lipids. “Older” RER then loses its connected ribosomes and is modified to become SER. Not everyone agrees with this interpretation, and other mechanisms of growth of ER are possible.

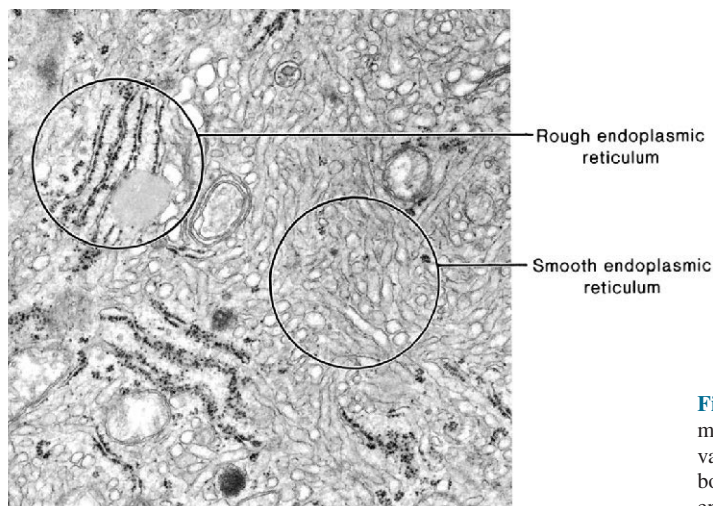
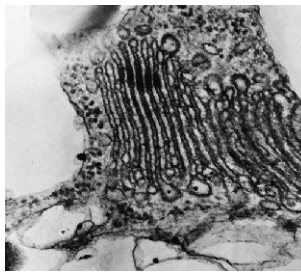
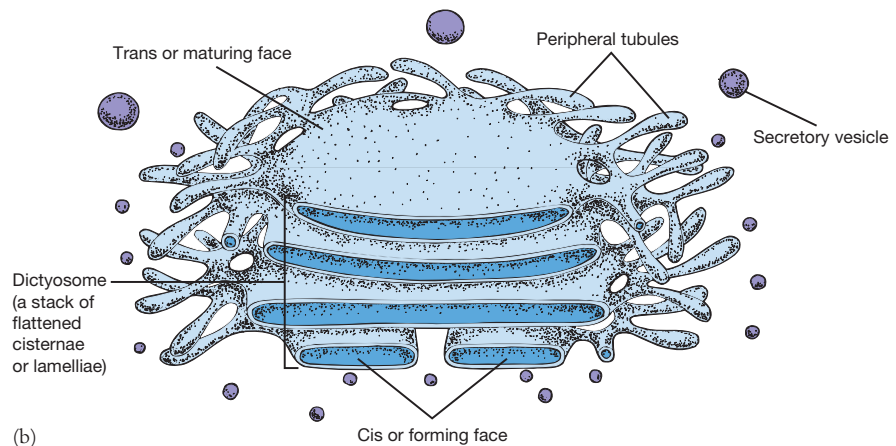


Figure 4.8 The Endoplasmic Reticulum. A transmission electron micrograph of the corpus luteum in a human ovary showing structural variations in eucaryotic endoplasmic reticulum. Note the presence of both rough endoplasmic reticulum lined with ribosomes and smooth endoplasmic reticulum without ribosomes ($\times 26,500$).



(a)

Figure 4.9 Golgi Apparatus Structure. Golgi apparatus of *Euglena gracilis*. Cisternal stacks are shown in the electron micrograph ($\times 165,000$) in (a) and diagrammatically in (b).



(b)

4.4 The Golgi Apparatus

The **Golgi apparatus** is a membranous organelle composed of flattened, saclike cisternae stacked on each other (**figure 4.9**). These membranes, like the smooth ER, lack bound ribosomes. There are usually around 4 to 8 cisternae or sacs in a stack, although there may be many more. Each sac is 15 to 20 nm thick and separated from other cisternae by 20 to 30 nm. A complex network of tubules and vesicles (20 to 100 nm in diameter) is located at the edges of the cisternae. The stack of cisternae has a definite polarity because there are two ends or faces that are quite different from one another. The sacs on the cis or forming face often are associated with the ER and differ from the sacs on the trans or maturing face in thickness, enzyme content, and degree of vesicle formation. It appears that material is transported from cis to trans cisternae by vesicles that bud off the cisternal edges and move to the next sac.

The Golgi apparatus is present in most eucaryotic cells, but many fungi and ciliate protozoa may lack a well-formed structure. Sometimes it consists of a single stack of cisternae; however, many cells may contain up to 20, and sometimes more, separate stacks. These stacks of cisternae, often called **dictyosomes**, can be clustered in one region or scattered about the cell.

The Golgi apparatus packages materials and prepares them for secretion, the exact nature of its role varying with the organism. The surface scales of some flagellated algae and radiolarian protozoa appear to be constructed within the Golgi apparatus and then transported to the surface in vesicles. It often participates in the development of cell membranes and in the packaging of cell products. The growth of some fungal hyphae occurs when Golgi vesicles contribute their contents to the wall at the hyphal tip.

In all these processes, materials move from the ER to the Golgi apparatus. Most often vesicles bud off the ER, travel to the Golgi apparatus, and fuse with the cis cisternae. Thus the Golgi apparatus is closely related to the ER in both a structural and a functional sense. Most proteins entering the Golgi apparatus from the ER are glycoproteins containing short carbohydrate chains. The Golgi apparatus frequently modifies proteins destined for

different fates by adding specific groups and then sends the proteins on their way to the proper location (e.g., lysosomal proteins have phosphates added to their mannose sugars).

4.5 Lysosomes and Endocytosis

A very important function of the Golgi apparatus and endoplasmic reticulum is the synthesis of another organelle, the **lysosome**. This organelle (or a structure very much like it) is found in a variety of microorganisms—protozoa, some algae, and fungi—as well as in plants and animals. Lysosomes are roughly spherical and enclosed in a single membrane; they average about 500 nm in diameter, but range from 50 nm to several μm in size. They are involved in intracellular digestion and contain the enzymes needed to digest all types of macromolecules. These enzymes, called hydrolases, catalyze the hydrolysis of molecules and function best under slightly acid conditions (usually around pH 3.5 to 5.0). Lysosomes maintain an acidic environment by pumping protons into their interior. Digestive enzymes are manufactured by the RER and packaged to form lysosomes by the Golgi apparatus. A segment of smooth ER near the Golgi apparatus also may bud off lysosomes.

Lysosomes are particularly important in those cells that obtain nutrients through **endocytosis**. In this process a cell takes up solutes or particles by enclosing them in vacuoles and vesicles pinched off from its plasma membrane. Vacuoles and vesicles are membrane-delimited cavities that contain fluid, and often solid material. Larger cavities will be called vacuoles, and smaller cavities, vesicles. There are two major forms of endocytosis: phagocytosis and pinocytosis. During **phagocytosis** large particles and even other microorganisms are enclosed in a phagocytic vacuole or phagosome and engulfed (**figure 4.10a**). In **pinocytosis** small amounts of the surrounding liquid with its solute molecules are pinched off as tiny pinocytotic vesicles (also called pinocytic vesicles) or pinosomes. Often phagosomes and pinosomes are collectively called **endosomes** because they are formed by endocytosis. The type of pinocytosis, receptor-mediated endocytosis, that produces coated vesicles (*see p. 403*) is important in the entry of animal viruses into host cells.

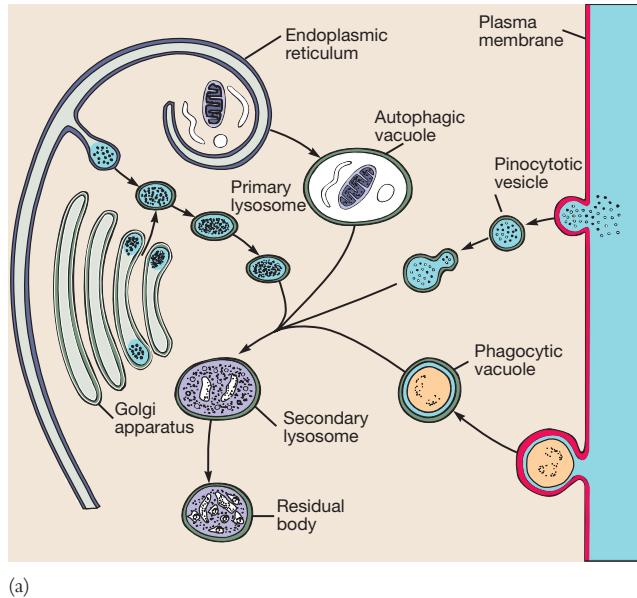
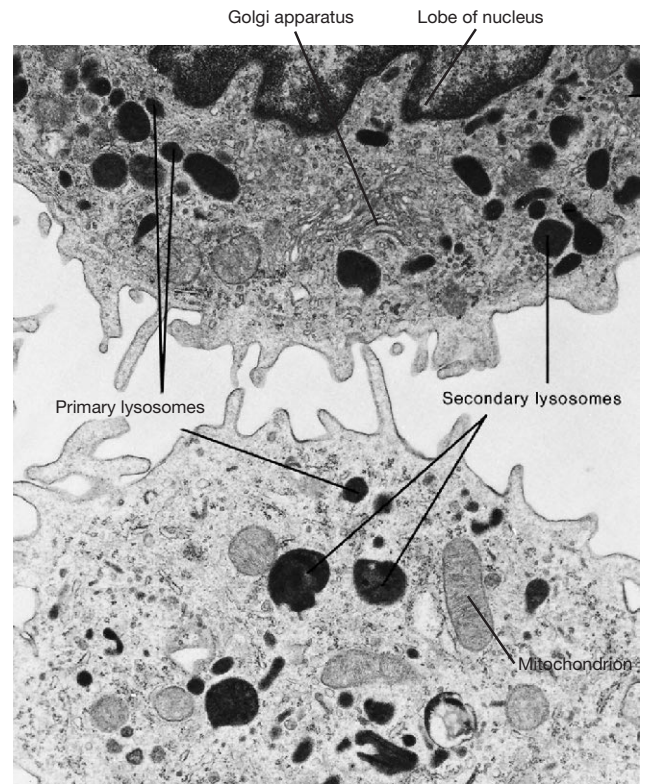


Figure 4.10 Lysosome Structure, Formation, and Function.

(a) A diagrammatic overview of lysosome formation and function. (b) Lysosomes in macrophages from the lung. Secondary lysosomes contain partially digested material and are formed by fusion of primary lysosomes and phagocytic vacuoles ($\times 14,137$).



Material in endosomes is digested with the aid of lysosomes. Newly formed lysosomes, or **primary lysosomes**, fuse with phagocytic vacuoles to yield **secondary lysosomes**, lysosomes with material being digested (figure 4.10). These phagocytic vacuoles or secondary lysosomes often are called food vacuoles. Digested nutrients then leave the secondary lysosome and enter the cytoplasm. When the lysosome has accumulated large quantities of indigestible material, it is known as a **residual body**.

Lysosomes join with phagosomes for defensive purposes as well as to acquire nutrients. Invading bacteria, ingested by a phagocytic cell, usually are destroyed when lysosomes fuse with the phagosome. This is commonly seen in leukocytes (white blood cells) of vertebrates. [Phagocytosis and resistance to pathogens \(pp. 718–20\)](#)

Cells can selectively digest portions of their own cytoplasm in a type of secondary lysosome called an **autophagic vacuole** (figure 4.10a). It is thought that these arise by lysosomal engulfment of a piece of cytoplasm (figure 4.11), or when the ER pinches off cytoplasm to form a vesicle that subsequently fuses with lysosomes. Autophagy probably plays a role in the normal turnover or recycling of cell constituents. A cell also can survive a period of starvation by selectively digesting portions of itself to remain alive. Following cell death, lysosomes aid in digestion and removal of cell debris.

A most remarkable thing about lysosomes is that they accomplish all these tasks without releasing their digestive enzymes into the cytoplasmic matrix, a catastrophe that would destroy the cell. The lysosomal membrane retains digestive enzymes and other macromolecules while allowing small digestion products to leave.

The intricate complex of membranous organelles composed of the Golgi apparatus, lysosomes, endosomes, and associated structures seems to operate as a coordinated whole whose main function is the import and export of materials (figure 4.11). Christian de Duve (Nobel Prize, 1974) has suggested that this complex be called the **vacuome** in recognition of its functional unity. The ER manufactures secretory proteins and membrane, and contributes these to the Golgi apparatus. The Golgi apparatus then forms secretory vesicles that fuse with the plasma membrane and release material to the outside. It also produces lysosomes that fuse with endosomes to digest material acquired through phagocytosis and pinocytosis. Membrane movement in the region of the vacuome lying between the Golgi apparatus and the plasma membrane is two-way. Empty vesicles often are recycled and returned to the Golgi apparatus and plasma membrane rather than being destroyed. These exchanges in the vacuome occur without membrane rupture so that vesicle contents never escape directly into the cytoplasmic matrix.

Figure 4.11 Membrane Flow in the Vacuome. The flow of material and membranes between organelles in a eucaryotic cell.

- (1) Vesicles shuttling between the ER and Golgi apparatus.
- (2) The Golgi-plasma membrane shuttle for secretion of materials.
- (3) The Golgi-lysosome shuttle.
- (4) The movement of material and membranes during endocytosis.
- (5) Pathways of plasma membrane recovery from endosomes, lysosomes, and through the Golgi apparatus.
- (6) Movement of vesicles from endosomes to lysosomes.
- (7) Autophagy by a lysosome.

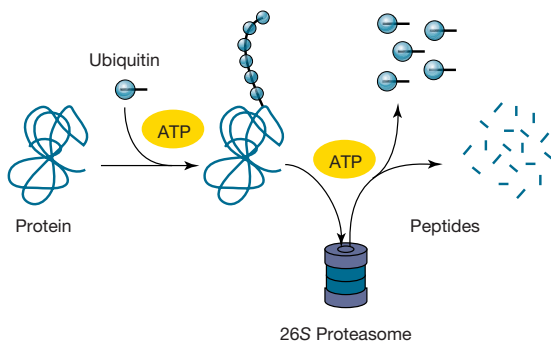
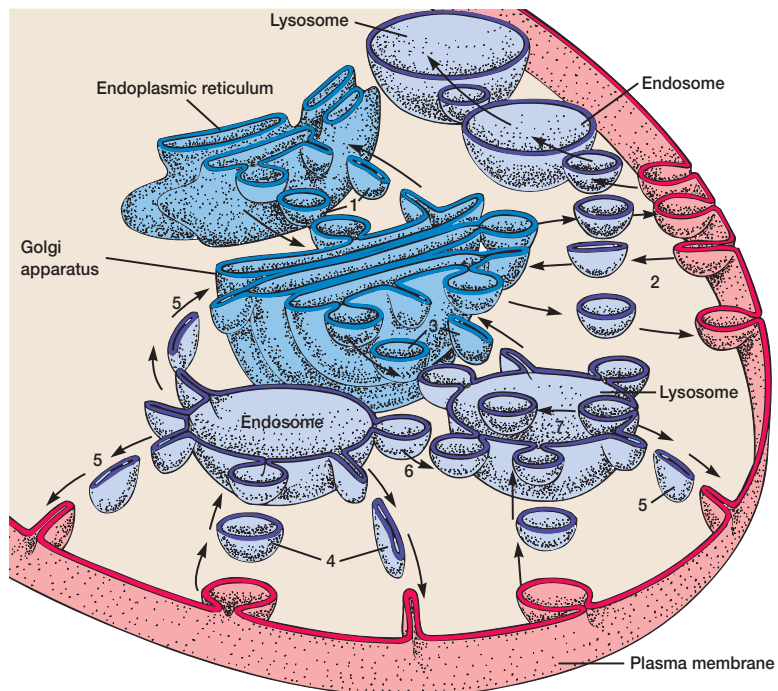


Figure 4.12 Proteasome Degradation of Proteins. See text for details.

More recently a nonlysosomal protein degradation system has been discovered in eucaryotic cells, a few bacteria, and many archaea. The majority of eucaryotic proteins may be degraded by this system. In eucaryotes, proteins are targeted for destruction by the attachment of several small ubiquitin polypeptides (**figure 4.12**). The marked protein then enters a huge cylindrical complex called a **26S proteasome**, where it is degraded to peptides in an

ATP-dependent process and the ubiquitins are released. The peptides may be hydrolyzed to amino acids. In this case the system is being used to recycle proteins. The proteasome also is involved in producing peptides for antigen presentation during many immunological responses (*see section 32.4*).

1. How do the rough and smooth endoplasmic reticulum differ from one another in terms of structure and function? List the processes in which the ER is involved.
2. Describe the structure of a Golgi apparatus in words and with a diagram. How do the cis and trans faces of the Golgi apparatus differ? List the major Golgi apparatus functions discussed in the text.
3. How are lysosomes formed? Describe the various forms of lysosomes and the way in which they participate in intracellular digestion. What is an autophagic vacuole? Define endocytosis, pinocytosis, and phagocytosis. What is a proteasome?

4.6 Eucaryotic Ribosomes

The eucaryotic ribosome can either be associated with the endoplasmic reticulum or be free in the cytoplasmic matrix and is larger than the bacterial 70S ribosome. It is a dimer of a 60S and

a 40S subunit, about 22 nm in diameter, and has a sedimentation coefficient of 80S and a molecular weight of 4 million. When bound to the endoplasmic reticulum to form rough ER, it is attached through its 60S subunit.

Both free and RER-bound ribosomes synthesize proteins. As mentioned earlier, proteins made on the ribosomes of the RER either enter its lumen for transport, and often for secretion, or are inserted into the ER membrane as integral membrane proteins. Free ribosomes are the sites of synthesis for nonsecretory and nonmembrane proteins. Some proteins synthesized by free ribosomes are inserted into organelles such as the nucleus, mitochondrion, and chloroplast. As discussed in chapters 3 and 12 (*see pp. 52, 272–74*), molecular chaperones aid the proper folding of proteins after synthesis. They also assist the transport of proteins into eucaryotic organelles such as mitochondria. Several ribosomes usually attach to a single messenger RNA and simultaneously translate its message into protein. These complexes of messenger RNA and ribosomes are called **polyribosomes** or **polysomes**. Ribosomal participation in protein synthesis is dealt with later. [The role of ribosomes in protein synthesis \(pp. 267–72\)](#)

1. Describe the structure of the eucaryotic 80S ribosome and contrast it with the procaryotic ribosome.
2. How do free ribosomes and those bound to the ER differ in function?

4.7 Mitochondria

Found in most eucaryotic cells, **mitochondria** (s., **mitochondrion**) frequently are called the “powerhouses” of the cell. Tricarboxylic acid cycle activity and the generation of ATP by electron transport and oxidative phosphorylation take place here. In the transmission electron microscope, mitochondria usually are cylindrical structures and measure approximately 0.3 to 1.0 μm by 5 to 10 μm . (In other words, they are about the same size as bacterial cells.) Although cells can possess as many as 1,000 or more mitochondria, at least a few cells (some yeasts, unicellular algae, and trypanosome protozoa) have a single giant tubular mitochondrion twisted into a continuous network permeating the cytoplasm (**figure 4.13**). [The tricarboxylic acid cycle, electron transport, and oxidative phosphorylation \(pp. 183–89\)](#)

The mitochondrion is bounded by two membranes, an outer mitochondrial membrane separated from an inner mitochondrial membrane by a 6 to 8 nm intermembrane space (**figure 4.14**). Special infoldings of the inner membrane, called **cristae** (s., **crista**), greatly increase its surface area. Their shape differs in mitochondria from various species. Fungi have platelike (laminar) cristae, whereas euglenoid flagellates may have cristae shaped like disks. Tubular cristae are found in a variety of eucaryotes; however,

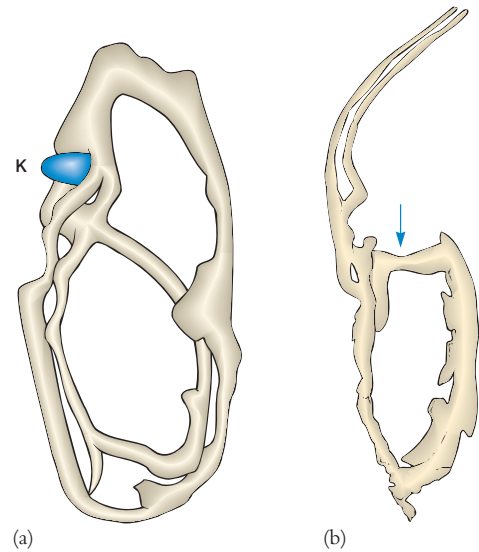


Figure 4.13 Trypanosome Mitochondria. The giant mitochondria from trypanosomes. **(a)** *Crithidia fasciculata* mitochondrion with kinetoplast, K. The kinetoplast contains DNA that codes for mitochondrial RNA and protein. **(b)** *Trypanosoma cruzi* mitochondrion with arrow indicating position of kinetoplast.

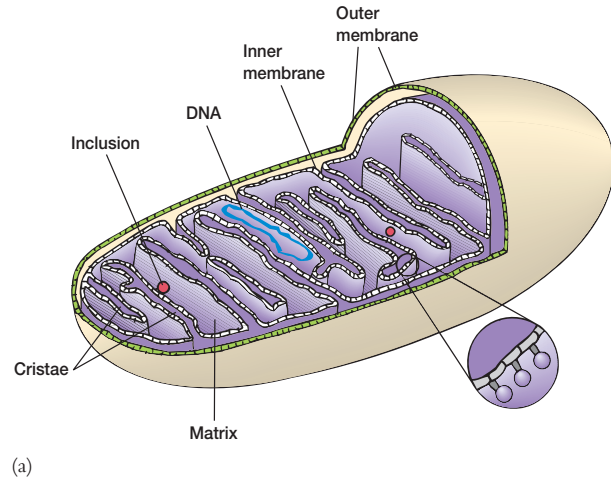
amoebae can possess mitochondria with cristae in the shape of vesicles (**figure 4.15**). The inner membrane encloses the mitochondrial matrix, a dense matrix containing ribosomes, DNA, and often large calcium phosphate granules. Mitochondrial ribosomes are smaller than cytoplasmic ribosomes and resemble those of bacteria in several ways, including their size and subunit composition. Mitochondrial DNA is a closed circle like bacterial DNA.

Each mitochondrial compartment is different from the others in chemical and enzymatic composition. The outer and inner mitochondrial membranes, for example, possess different lipids. Enzymes and electron carriers involved in electron transport and oxidative phosphorylation (the formation of ATP as a consequence of electron transport) are located only in the inner membrane. The enzymes of the tricarboxylic acid cycle and the β -oxidation pathway for fatty acids (*see chapter 9*) are located in the matrix.

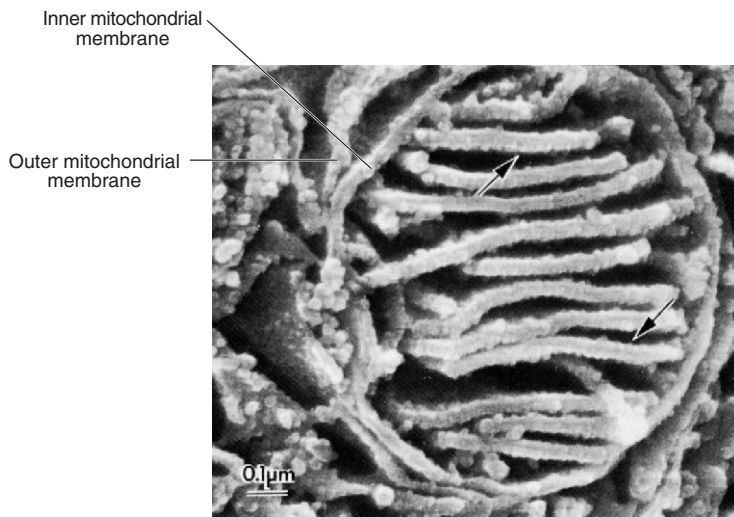
The inner membrane of the mitochondrion has another distinctive structural feature related to its function. Many small spheres, about 8.5 nm diameter, are attached by stalks to its inner surface. The spheres are called **F₁ particles** and synthesize ATP during cellular respiration (*see pp. 187–89*).

The mitochondrion uses its DNA and ribosomes to synthesize some of its own proteins. In fact, mutations in mitochondrial DNA often lead to serious diseases in humans. Most mitochondrial proteins, however, are manufactured under the direction of

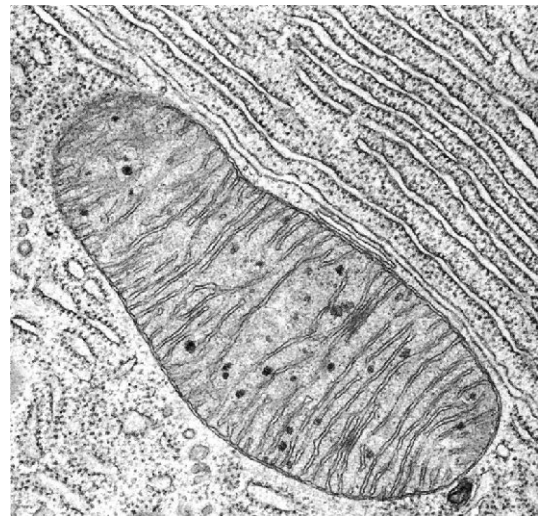
Figure 4.14 Mitochondrial Structure. (a) A diagram of mitochondrial structure. The insert shows F_1F_0 complexes lining the inner surface of the cristae. (b) Scanning electron micrograph ($\times 70,000$) of a freeze-fractured mitochondrion showing the cristae (arrows). The outer and inner mitochondrial membranes also are evident. (c) Transmission electron micrograph of a mitochondrion from a bat pancreas ($\times 85,000$). Note outer and inner mitochondrial membranes, cristae, and inclusions in the matrix. The mitochondrion is surrounded by rough endoplasmic reticulum.



(a)

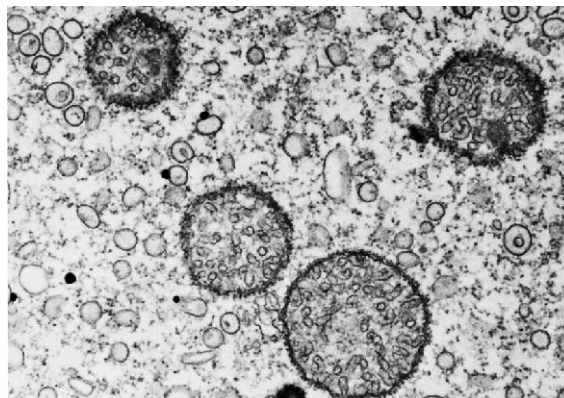


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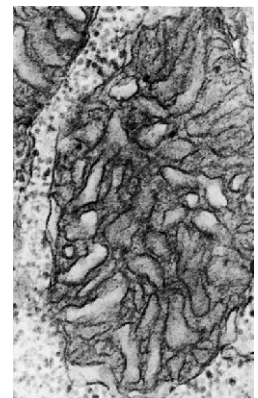


(c)

Figure 4.15 Mitochondrial Cristae. Mitochondria with a variety of cristae shapes. (a) Mitochondria from the protostelid slime mold *Schizoplasmodiopsis micropunctata*. Note the tubular cristae ($\times 49,500$). (b) The protozoan *Actinosphaerium* with vesicular cristae ($\times 75,000$).



(a)



(b)

Box 4.1

The Origin of the Eucaryotic Cell

The profound differences between eucaryotic and procaryotic cells have stimulated much discussion about how the more complex eucaryotic cell arose. Some biologists believe the original “protoeucaryote” was a large aerobic archaean or bacterium that formed mitochondria, chloroplasts, and nuclei when its plasma membrane invaginated and enclosed genetic material in a double membrane. The organelles could then evolve independently. It also is possible that a large blue-green bacterium lost its cell wall and became phagocytic. Subsequently, primitive chloroplasts, mitochondria, and nuclei would be formed by the fusion of thylakoids and endoplasmic reticulum cisternae to enclose specific areas of cytoplasm.

By far the most popular theory for the origin of eucaryotic cells is the **endosymbiotic theory**. In brief, it is supposed that the ancestral procaryotic cell, which may have been an archaean, lost its cell wall and gained the ability to obtain nutrients by phagocytosing other procaryotes. When photosynthetic cyanobacteria arose, the environment slowly became aerobic. If an anaerobic, amoeboid, phagocytic procaryote—possibly already possessing a developed nucleus—engulfed an aerobic bacterial cell and established a permanent symbiotic relationship with it, the host would be better adapted to its increasingly aerobic environment. The endosymbiotic aerobic bacterium eventually would develop into the mitochondrion. Similarly, symbiotic associations with cyanobacteria could lead to the forma-

tion of chloroplasts and photosynthetic eucaryotes. Some have speculated that cilia and flagella might have arisen from the attachment of spirochete bacteria (*see chapter 21*) to the surface of eucaryotic cells, much as spirochetes attach themselves to the surface of the motile protozoan *Myxotricha paradoxa* that grows in the digestive tract of termites.

There is evidence to support the endosymbiotic theory. Both mitochondria and chloroplasts resemble bacteria in size and appearance, contain DNA in the form of a closed circle like that of bacteria, and reproduce semiautonomously. Mitochondrial and chloroplast ribosomes resemble procaryotic ribosomes more closely than those in the eucaryotic cytoplasmic matrix. The sequences of the chloroplast and mitochondrial genes for ribosomal RNA and transfer RNA are more similar to bacterial gene sequences than to those of eucaryotic rRNA and tRNA nuclear genes. Finally, there are symbiotic associations that appear to be bacterial endosymbioses in which distinctive procaryotic characteristics are being lost. For example, the protozoan flagellate *Cyanophora paradoxa* has photosynthetic organelles called cyanellae with a structure similar to that of cyanobacteria and the remains of peptidoglycan in their walls. Their DNA is much smaller than that of cyanobacteria and resembles chloroplast DNA. Despite such evidence, the endosymbiotic theory still is somewhat speculative and the center of much continuing research and discussion.

the nucleus. Mitochondria reproduce by binary fission. Chloroplasts show similar partial independence and reproduction by binary fission. Because both organelles resemble bacteria to some extent, it has been suggested that these organelles arose from symbiotic associations between bacteria and larger cells (**Box 4.1**).

4.8 Chloroplasts

Plastids are cytoplasmic organelles of algae and higher plants that often possess pigments such as chlorophylls and carotenoids, and are the sites of synthesis and storage of food reserves. The most important type of plastid is the chloroplast. **Chloroplasts** contain chlorophyll and use light energy to convert CO₂ and water to carbohydrates and O₂. That is, they are the site of photosynthesis.

Although chloroplasts are quite variable in size and shape, they share many structural features. Most often they are oval with dimensions of 2 to 4 μm by 5 to 10 μm, but some algae possess one huge chloroplast that fills much of the cell. Like mitochondria, chloroplasts are encompassed by two membranes (**figure 4.16**). A matrix, the **stroma**, lies within the inner membrane. It contains DNA, ribosomes, lipid droplets, starch granules, and a complex internal membrane system whose most prominent components are

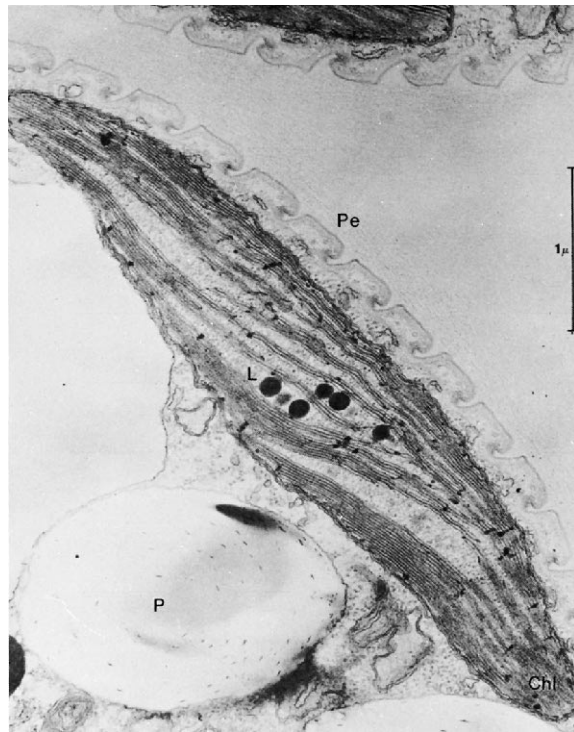
flattened, membrane-delimited sacs, the **thylakoids**. Clusters of two or more thylakoids are dispersed within the stroma of most algal chloroplasts (**figures 4.16 and 4.25b**). In some groups of algae, several disklike thylakoids are stacked on each other like coins to form **grana** (s., **granum**).

Photosynthetic reactions are separated structurally in the chloroplast just as electron transport and the tricarboxylic acid cycle are in the mitochondrion. The formation of carbohydrate from CO₂ and water, the dark reaction, takes place in the stroma. The trapping of light energy to generate ATP, NADPH, and O₂, the light reaction, is located in the thylakoid membranes, where chlorophyll and electron transport components are also found.

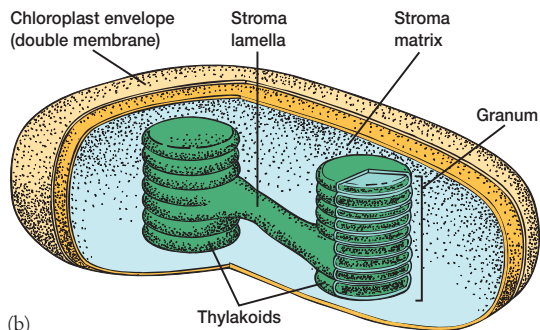
[Photosynthesis \(pp. 195–201\)](#)

The chloroplasts of many algae contain a **pyrenoid** (**figure 4.25b**), a dense region of protein surrounded by starch or another polysaccharide. Pyrenoids participate in polysaccharide synthesis.

1. Describe in detail the structure of mitochondria and chloroplasts. Where are the different components of these organelles' energy trapping systems located?
2. Define F₁ particle, plastid, dark reaction, light reaction, and pyrenoid.
3. What is the role of mitochondrial DNA?



(a)



(b)

Figure 4.16 Chloroplast Structure. (a) The chloroplast (Chl), of the euglenoid flagellate *Colacium cyclopicolum*. The chloroplast is bounded by a double membrane and has its thylakoids in groups of three or more. A paramylon granule (P), lipid droplets (L), and the pellicular strips (Pe), can be seen ($\times 40,000$). (b) A diagram of chloroplast structure.

4.9 The Nucleus and Cell Division

The cell **nucleus** is by far the most visually prominent organelle. It was discovered early in the study of cell structure and was shown by Robert Brown in 1831 to be a constant feature of eucaryotic cells. The nucleus is the repository for the cell's genetic information and is its control center.

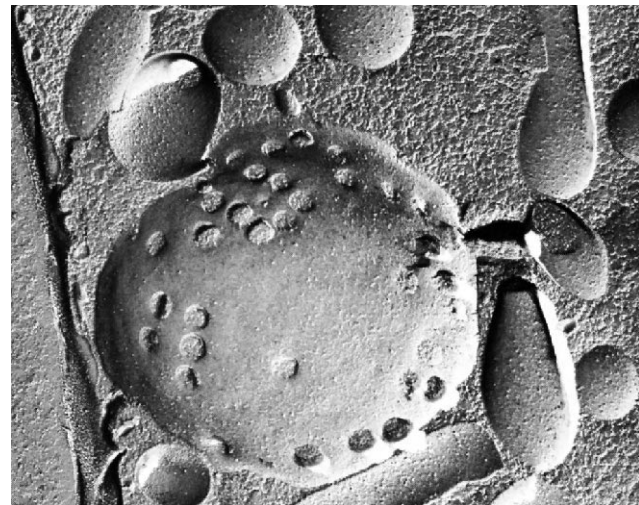


Figure 4.17 The Nucleus. A freeze-etch preparation of the conidium of the fungus *Geotrichum candidum* ($\times 44,600$). Note the large convex nuclear surface with nuclear pores scattered over it.

Nuclear Structure

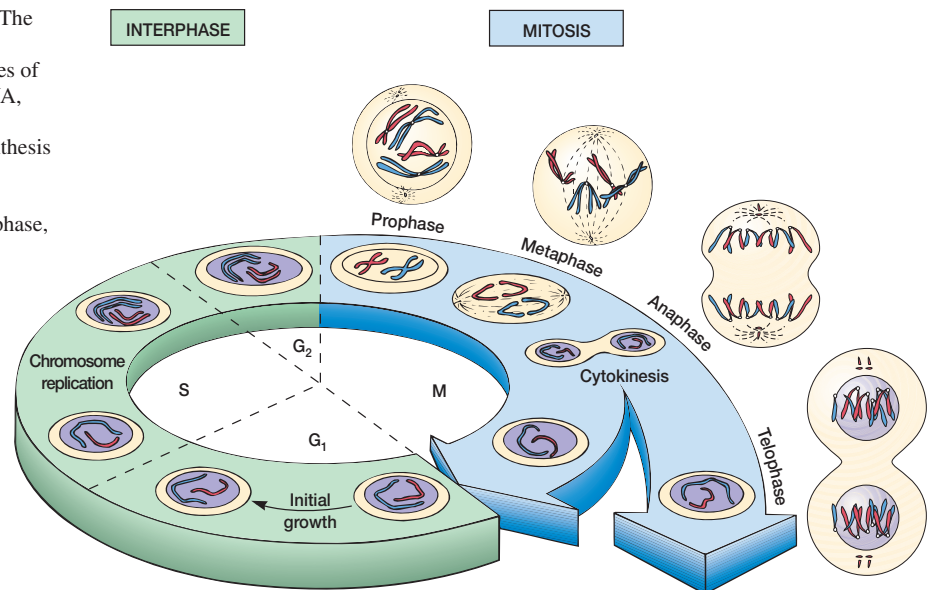
Nuclei are membrane-delimited spherical bodies about 5 to 7 μm in diameter (figures 4.2 and 4.25b). Dense fibrous material called **chromatin** can be seen within the nucleoplasm of the nucleus of a stained cell. This is the DNA-containing part of the nucleus. In nondividing cells, chromatin exists in a dispersed condition, but condenses during mitosis to become visible as **chromosomes**. Some nuclear chromatin, the euchromatin, is loosely organized and contains those genes that are expressing themselves actively. Heterochromatin is coiled more tightly, appears darker in the electron microscope, and is not genetically active most of the time. [Organization of DNA in eucaryotic nuclei \(pp. 234–35\)](#)

The nucleus is bounded by the **nuclear envelope** (figures 4.2 and 4.25b), a complex structure consisting of inner and outer membranes separated by a 15 to 75 nm perinuclear space. The envelope is continuous with the ER at several points and its outer membrane is covered with ribosomes. A network of intermediate filaments, called the nuclear lamina, lies against the inner surface of the envelope and supports it. Chromatin usually is associated with the inner membrane.

Many **nuclear pores** penetrate the envelope (**figure 4.17**), each pore formed by a fusion of the outer and inner membranes. Pores are about 70 nm in diameter and collectively occupy about 10 to 25% of the nuclear surface. A complex ringlike arrangement of granular and fibrous material called the annulus is located at the edge of each pore.

The nuclear pores serve as a transport route between the nucleus and surrounding cytoplasm. Particles have been observed moving into the nucleus through the pores. Although the function of the annulus is not understood, it may either regulate or aid the movement of material through the pores. Substances also move directly through the nuclear envelope by unknown mechanisms.

Figure 4.18 The Eucaryotic Cell Cycle. The length of the M period has been increased disproportionately in order to show the phases of mitosis. G_1 period: synthesis of mRNA, tRNA, ribosomes, and cytoplasmic constituents. Nucleolus grows rapidly. S period: rapid synthesis and doubling of nuclear DNA and histones. G_2 period: preparation for mitosis and cell division. M period: mitosis (prophase, metaphase, anaphase, telophase) and cytokinesis.



The Nucleolus

Often the most noticeable structure within the nucleus is the **nucleolus** (figures 4.2 and 4.25*b*). A nucleus may contain from one to many nucleoli. Although the nucleolus is not membrane-enclosed, it is a complex organelle with separate granular and fibrillar regions. It is present in nondividing cells, but frequently disappears during mitosis. After mitosis the nucleolus reforms around the nucleolar organizer, a particular part of a specific chromosome.

The nucleolus plays a major role in ribosome synthesis. The nucleolar organizer DNA directs the production of ribosomal RNA (rRNA). This RNA is synthesized in a single long piece that then is cut to form the final rRNA molecules. The processed rRNAs next combine with ribosomal proteins (which have been synthesized in the cytoplasmic matrix) to form partially completed ribosomal subunits. The granules seen in the nucleolus are probably these subunits. Immature ribosomal subunits then leave the nucleus, presumably by way of the nuclear envelope pores and mature in the cytoplasm. [RNA splicing \(p. 264\)](#)

Mitosis and Meiosis

When a eucaryotic microorganism reproduces, its genetic material must be duplicated and then separated so that each new nucleus possesses a complete set of chromosomes. This process of nuclear division and chromosome distribution in eucaryotic cells is called **mitosis**. Mitosis actually occupies only a small portion of a microorganism's life as can be seen by examining the **cell cycle** (figure 4.18). The cell cycle is the total sequence of events in the growth-division cycle between the end of one division and the end of the next. Cell growth takes place in the **interphase**, that portion of the cycle between periods of mitosis. Interphase is composed of three parts. The G_1 period (gap 1 period) is a time

of active synthesis of RNA, ribosomes, and other cytoplasmic constituents accompanied by considerable cell growth. This is followed by the S period (synthesis period) in which DNA is replicated and doubles in quantity. Finally, there is a second gap, the G_2 period, when the cell prepares for mitosis, the M period, by activities such as the synthesis of special division proteins. The total length of the cycle differs considerably between microorganisms, usually due to variations in the length of G_1 .

Mitotic events are summarized in figure 4.18. During mitosis, the genetic material duplicated during the S period is distributed equally to the two new nuclei so that each has a full set of genes. There are four phases in mitosis. In prophase, the chromosomes—each with two chromatids—become visible and move toward the equator of the cell. The mitotic spindle forms, the nucleolus disappears, and the nuclear envelope begins to dissolve. The chromosomes are arranged in the center of the spindle during metaphase and the nuclear envelope has disappeared. During anaphase the chromatids in each chromosome separate and move toward the opposite poles of the spindle. Finally during telophase the chromatids become less visible, the nucleolus reappears, and a nuclear envelope reassembles around each set of chromatids to form two new nuclei.

Mitosis in eucaryotic microorganisms can differ from that pictured in figure 4.18. For example, the nuclear envelope does not disappear in many fungi and some protozoa and algae (figure 4.19). Frequently cytokinesis, the division of the parental cell's cytoplasm to form new cells, begins during anaphase and finishes by the end of telophase. However, mitosis can take place without cytokinesis to generate multinucleate or coenocytic cells.

In mitosis the original number of chromosomes is the same after division and a diploid organism will remain diploid or $2N$ (i.e., it still has two copies of each chromosome). Frequently a microorganism reduces its chromosome number by half, from the diploid state to the haploid or $1N$ (a single copy of each chromosome).

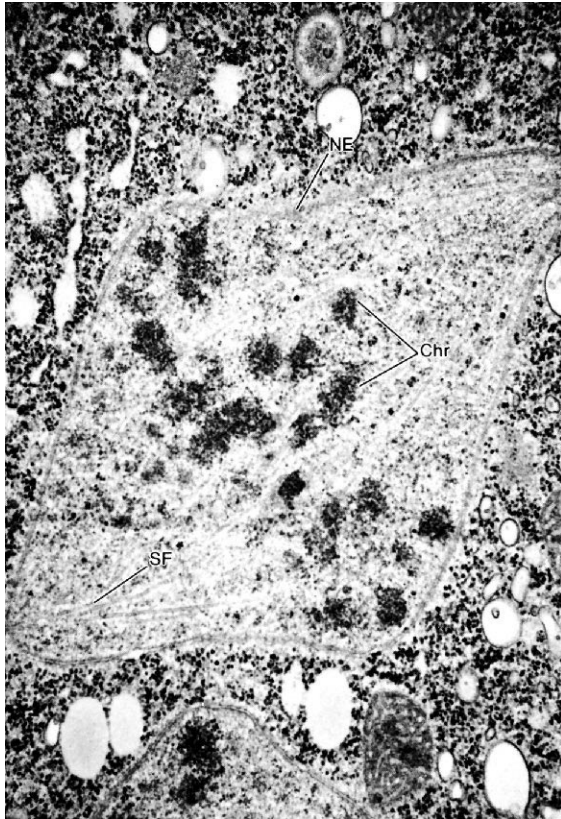


Figure 4.19 Mitosis with an Intact Nuclear Envelope. Mitosis in the slime mold *Physarum flavicomum*. The nuclear envelope, NE, remains intact, and the spindle is intranuclear. The process is at metaphase with the chromosomes, Chr, aligned in the center and attached to spindle fibers, SF ($\times 15,000$).

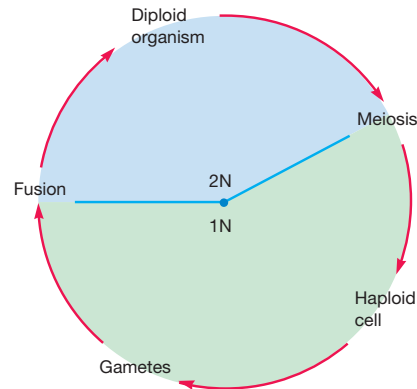


Figure 4.20 Generalized Eucaryotic Life Cycle.

single-stranded chromosomes are separated. After completion of meiosis I and meiosis II, the original diploid cell has been transformed into four haploid cells.

1. Describe the structure of the nucleus. What are euchromatin and heterochromatin? What is the role of the pores in the nuclear envelope?
2. Briefly discuss the structure and function of the nucleolus. What is the nucleolar organizer?
3. Describe the eucaryotic cell cycle, its periods, and the process of mitosis. What is meiosis, how does it take place, and what is its role in the microbial life cycle?

Haploid cells may immediately act as gametes and fuse to reform diploid organisms or may form gametes only after a considerable delay (**figure 4.20**). The process by which the number of chromosomes is reduced in half with each daughter cell receiving one complete set of chromosomes is called **meiosis**. Life cycles can be quite complex in eucaryotic microorganisms; a classic example is the life cycle of *Plasmodium*, the cause of malaria (*see pp. 954–56*). [Life cycles of eucaryotic microorganisms \(chapters 25–27\)](#)

Meiosis is quite complex and involves two stages. The first stage differs markedly from mitosis. During prophase, homologous chromosomes come together and lie side-by-side, a process known as synapsis. Then the double-stranded chromosomes from each homologous pair move to opposite poles in anaphase. In contrast, during mitotic anaphase the two strands of each chromosome separate and move to opposite poles. Consequently the number of chromosomes is halved in meiosis but not in mitosis. The second stage of meiosis is similar to mitosis in terms of mechanics, and

4.10 External Cell Coverings

Eucaryotic microorganisms differ greatly from procaryotes in the supporting or protective structures they have external to the plasma membrane. In contrast with most bacteria, many eucaryotes lack an external cell wall. The amoeba is an excellent example. Eucaryotic cell membranes, unlike most procaryotic membranes, contain sterols such as cholesterol in their lipid bilayers, and this may make them mechanically stronger, thus reducing the need for external support. (However, as mentioned on page 47, many procaryotic membranes are strengthened by hopanoids.) Of course many eucaryotes do have a rigid external **cell wall**. Algal cell walls usually have a layered appearance and contain large quantities of polysaccharides such as cellulose and pectin. In addition, inorganic substances like silica (in diatoms) or calcium carbonate (some red algae) may be present. Fungal cell walls normally are rigid. Their exact composition varies with the organism; but usually, cellulose, chitin, or glucan (a glucose polymer different from cellulose) are present. Despite their nature the rigid materials in eucaryotic walls

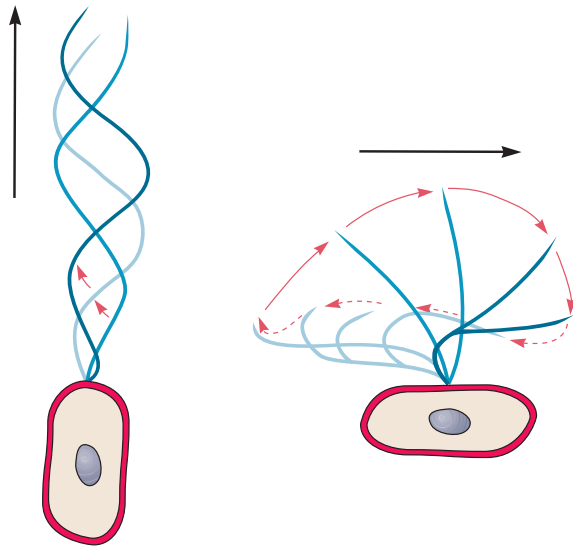


Figure 4.21 Patterns of Flagellar Movement. Flagellar movement (left illustration) often takes the form of waves that move either from the base of the flagellum to its tip or in the opposite direction. The motion of these waves propels the organism along. The beat of a cilium (right illustration) may be divided into two phases. In the effective stroke, the cilium remains fairly stiff as it swings through the water. This is followed by a recovery stroke in which the cilium bends and returns to its initial position. The black arrows indicate the direction of water movement in these examples.

are chemically simpler than procaryotic peptidoglycan. [Bacterial cell wall structure and chemistry \(pp. 55–60\)](#)

Many protozoa and some algae have a different external structure, the **pellicle** (figure 4.16a). This is a relatively rigid layer of components just beneath the plasma membrane (sometimes the plasma membrane is also considered part of the pellicle). The pellicle may be fairly simple in structure. For example, *Euglena* has a series of overlapping strips with a ridge at the edge of each strip fitting into a groove on the adjacent one. In contrast, ciliate protozoan pellicles are exceptionally complex with two membranes and a variety of associated structures. Although pellicles are not as strong and rigid as cell walls, they do give their possessors a characteristic shape.

4.11 Cilia and Flagella

Cilia (s., **cilium**) and **flagella** (s., **flagellum**) are the most prominent organelles associated with motility. Although both are whiplike and beat to move the microorganism along, they differ from one another in two ways. First, cilia are typically only 5 to 20 μm in length, whereas flagella are 100 to 200 μm long. Second, their patterns of movement are usually distinctive (figure 4.21). Fla-

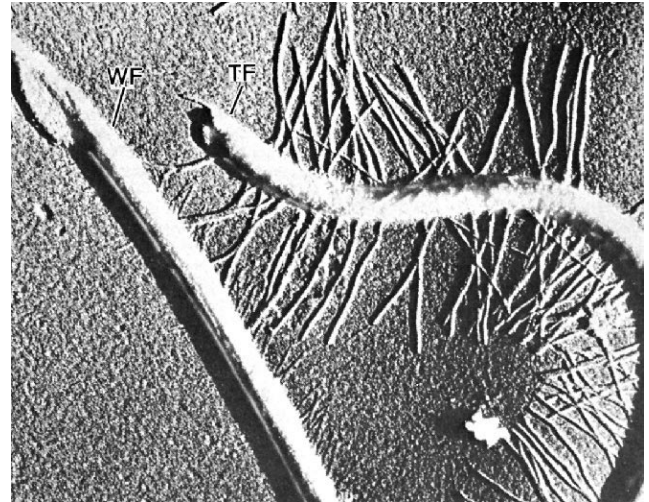


Figure 4.22 Whiplash and Tinsel Flagella. Transmission electron micrograph of a shadowed whiplash flagellum, WF, and a tinsel flagellum, TF, with mastigonemes.

gella move in an undulating fashion and generate planar or helical waves originating at either the base or the tip. If the wave moves from base to tip, the cell is pushed along; a beat traveling from the tip toward the base pulls the cell through the water. Sometimes the flagellum will have lateral hairs called flimmer filaments (thicker, stiffer hairs are called mastigonemes). These filaments change flagellar action so that a wave moving down the filament toward the tip pulls the cell along instead of pushing it. Such a flagellum often is called a tinsel flagellum, whereas the naked flagellum is referred to as a whiplash flagellum (figure 4.22). Cilia, on the other hand, normally have a beat with two distinctive phases. In the effective stroke, the cilium strokes through the surrounding fluid like an oar, thereby propelling the organism along in the water. The cilium next bends along its length while it is pulled forward during the recovery stroke in preparation for another effective stroke. A ciliated microorganism actually coordinates the beats so that some of its cilia are in the recovery phase while others are carrying out their effective stroke (figure 4.23). This coordination allows the organism to move smoothly through the water.

Despite their differences, cilia and flagella are very similar in ultrastructure. They are membrane-bound cylinders about 0.2 μm in diameter. Located in the matrix of the organelle is a complex, the **axoneme**, consisting of nine pairs of microtubule doublets arranged in a circle around two central tubules (figure 4.24). This is called the 9 + 2 pattern of microtubules. Each doublet also has pairs of arms projecting from subtubule A (the complete microtubule) toward a neighboring doublet. A radial spoke extends from subtubule A toward the internal pair of microtubules with their central sheath. These microtubules are similar to those found in the cytoplasm. Each is constructed of two types



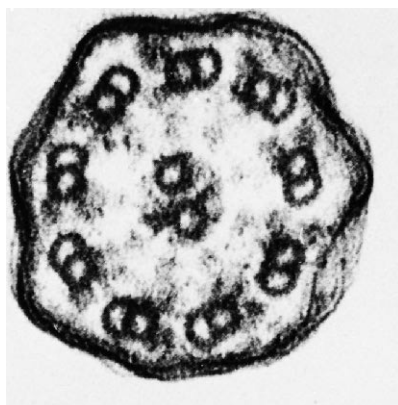
Figure 4.23 Coordination of Ciliary Activity. A scanning electron micrograph of *Paramecium* showing cilia ($\times 1,500$). The ciliary beat is coordinated and moves in waves across the protozoan's surface, as can be seen in the photograph.

of tubulin subunits, α - and β -tubulins, that resemble the contractile protein actin in their composition. [Bacterial flagella and motility \(pp. 63–66\)](#)

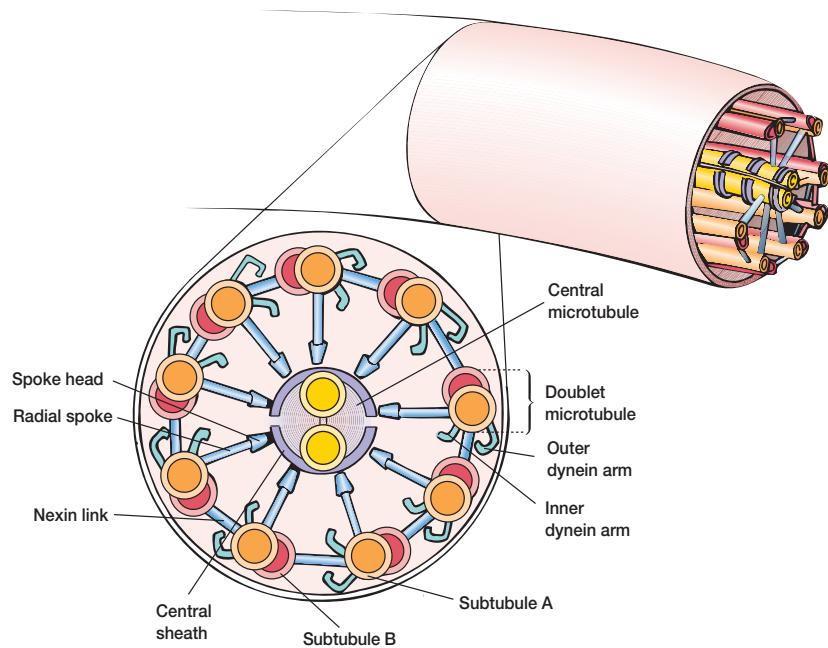
A **basal body** lies in the cytoplasm at the base of each cilium or flagellum. It is a short cylinder with nine microtubule triplets around its periphery (a 9 + 0 pattern) and is separated from the rest of the organelle by a basal plate. The basal body directs the construction of these organelles. Cilia and flagella appear to grow through the addition of preformed microtubule subunits at their tips.

Cilia and flagella bend because adjacent microtubule doublets slide along one another while maintaining their individual lengths. The doublet arms (figure 4.24), about 15 nm long, are made of the protein **dynein**. ATP powers the movement of cilia and flagella, and isolated dynein hydrolyzes ATP. It appears that dynein arms interact with the B subtubules of adjacent doublets to cause the sliding. The radial spokes also participate in this sliding motion.

Cilia and flagella beat at a rate of about 10 to 40 strokes or waves per second and propel microorganisms rapidly. The record holder is the flagellate *Monas stigmatica*, which swims at a rate of 260 $\mu\text{m}/\text{second}$ (approximately 40 cell lengths per second); the common euglenoid flagellate, *Euglena gracilis*, travels at around 170 μm or 3 cell lengths per second. The ciliate protozoan *Paramecium caudatum* swims at about 2,700 $\mu\text{m}/\text{second}$ (12 lengths per second). Such speeds are equivalent to or much faster than those seen in higher animals.



(a)



(b)

Figure 4.24 Cilia and Flagella Structure. (a) An electron micrograph of a cilium cross section. Note the two central microtubules surrounded by nine microtubule doublets ($\times 160,000$). (b) A diagram of cilia and flagella structure with two doublets removed for sake of visibility.

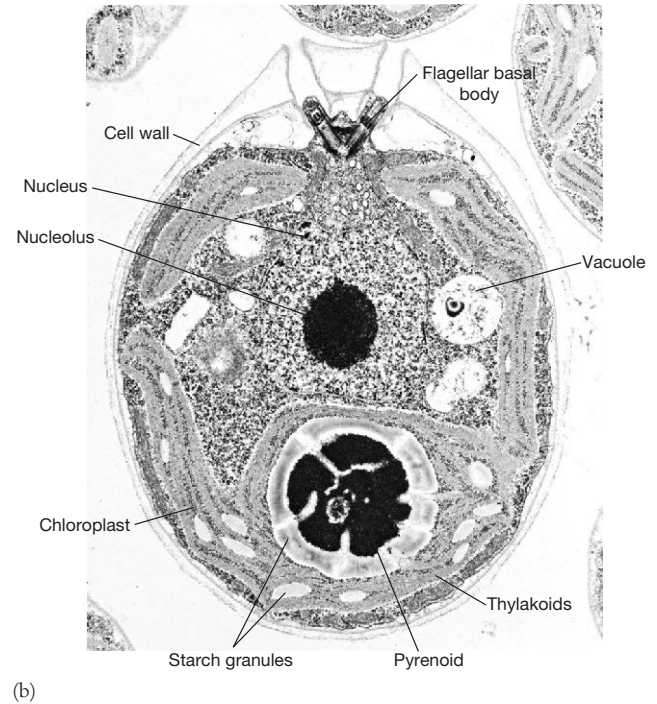
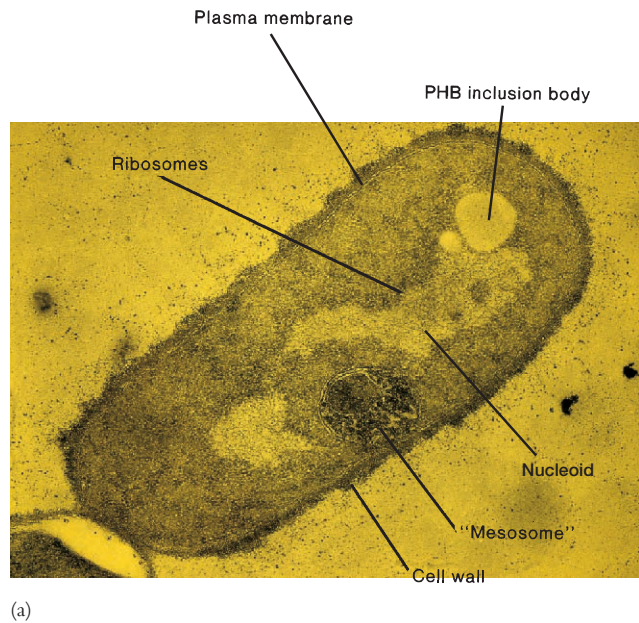


Figure 4.25 Comparison of Prokaryotic and Eucaryotic Cell Structure. (a) The prokaryote *Bacillus megaterium* ($\times 30,500$). (b) The eucaryotic alga *Chlamydomonas reinhardtii*, a deflagellated cell. Note the large chloroplast with its pyrenoid body ($\times 30,000$).

1. How do eucaryotic microorganisms differ from procaryotes with respect to supporting or protective structures external to the plasma membrane? Describe the pellicle and indicate which microorganisms have one.
2. Prepare and label a diagram showing the detailed structure of a cilium or flagellum. How do cilia and flagella move, and what is dynein's role in the process?

4.12 Comparison of Prokaryotic and Eucaryotic Cells

A comparison of the cells in **figure 4.25** demonstrates that there are many fundamental differences between eucaryotic and procaryotic cells. **Eucaryotic cells** have a membrane-enclosed nucleus. In contrast, **procaryotic cells** lack a true, membrane-delimited nucleus. Bacteria and Archaea are procaryotes; all other organisms—algae, fungi, protozoa, higher plants, and animals—are eucaryotic. Procaryotes normally are smaller than eucaryotic cells, often about the size of eucaryotic mitochondria and chloroplasts.

The presence of the eucaryotic nucleus is the most obvious difference between these two cell types, but several other major distinctions should be noted. It is clear from **table 4.2** that procaryotic cells are much simpler structurally. In particular, an extensive and diverse collection of membrane-delimited organelles is missing. Furthermore, procaryotes are simpler functionally in several ways. They lack mitosis and meiosis, and have a simpler genetic organization. Many complex eucaryotic processes are absent in procaryotes: phagocytosis and pinocytosis, intracellular digestion, directed cytoplasmic streaming, ameboid movement, and others.

Despite the many significant differences between these two basic cell forms, they are remarkably similar on the biochemical level as will be discussed in succeeding chapters. Procaryotes and eucaryotes are composed of similar chemical constituents. With a few exceptions the genetic code is the same in both, as is the way in which the genetic information in DNA is expressed. The principles underlying metabolic processes and most of the more important metabolic pathways are identical. Thus beneath the profound structural and functional differences between procaryotes and eucaryotes, there is an even more fundamental unity: a molecular unity that is basic to all known life processes.

Table 4.2 Comparison of Prokaryotic and Eucaryotic Cells

Property	Prokaryotes	Eucaryotes
Organization of Genetic Material		
True membrane-bound nucleus	Absent	Present
DNA complexed with histones	No	Yes
Number of chromosomes	One ^a	More than one
Introns in genes	Rare	Common
Nucleolus	Absent	Present
Mitosis occurs	No	Yes
Genetic Recombination	Partial, unidirectional transfer of DNA	Meiosis and fusion of gametes
Mitochondria	Absent	Present
Chloroplasts	Absent	Present
Plasma Membrane with Sterols	Usually no ^b	Yes
Flagella	Submicroscopic in size; composed of one fiber	Microscopic in size; membrane bound; usually 20 microtubules in 9 + 2 pattern
Endoplasmic Reticulum	Absent	Present
Golgi Apparatus	Absent	Present
Cell Walls	Usually chemically complex with peptidoglycan ^c	Chemically simpler and lacking peptidoglycan
Differences in Simpler Organelles		
Ribosomes	70S	80S (except in mitochondria and chloroplasts)
Lysosomes and peroxisomes	Absent	Present
Microtubules	Absent or rare	Present
Cytoskeleton	May be absent	Present
Differentiation	Rudimentary	Tissues and organs

^aPlasmids may provide additional genetic information.

^bOnly the mycoplasmas and methanotrophs (methane utilizers) contain sterols. The mycoplasmas cannot synthesize sterols and require them preformed. Many prokaryotes contain hopanoids.

^cThe mycoplasmas and Archaea do not have peptidoglycan cell walls.

Summary

- The eucaryotic cell has a true, membrane-delimited nucleus and many membranous organelles (**table 4.1**).
- The cytoplasmic matrix contains microfilaments, intermediate filaments, and microtubules, small organelles partly responsible for cell structure and movement. These and other types of filaments are organized into a cytoskeleton.
- The matrix is permeated by an irregular network of tubules and flattened sacs or cisternae known as the endoplasmic reticulum (ER). The ER may have attached ribosomes and be active in protein synthesis (rough or granular endoplasmic reticulum) or lack ribosomes (smooth or agranular ER).
- The ER can donate materials to the Golgi apparatus, an organelle composed of one or more stacks of cisternae (**figure 4.9**). This organelle prepares and packages cell products for secretion.
- The Golgi apparatus also forms lysosomes (**figures 4.10 and 4.11**). These organelles contain digestive enzymes and aid in intracellular digestion of materials, including those taken up by endocytosis.
- Eucaryotic ribosomes found free in the cytoplasmic matrix or bound to the ER are 80S ribosomes. Several may be attached to the same messenger RNA forming polyribosomes or polysomes.
- Mitochondria are organelles bounded by two membranes, with the inner membrane folded into cristae, and are responsible for energy generation by the tricarboxylic acid cycle, electron transport, and oxidative phosphorylation (**figure 4.14**).
- Chloroplasts are the site of photosynthesis. The trapping of light energy takes place in the thylakoid membranes, whereas CO₂ incorporation is located in the stroma (**figure 4.16**).
- The nucleus is a large organelle containing the cell's chromosomes. It is bounded by a complex, double-membrane envelope perforated by pores through which materials can move.
- The nucleolus lies within the nucleus and participates in the synthesis of ribosomal RNA and ribosomal subunits.
- Eucaryotic chromosomes are distributed to daughter cells during regular cell division by mitosis (**figure 4.18**). Meiosis is used to halve the chromosome number during sexual reproduction.
- When a cell wall is present, it is constructed from polysaccharides, like cellulose, that are chemically simpler than prokaryotic peptidoglycan. Many protozoa have a pellicle rather than a cell wall.
- Many eucaryotic cells are motile because of cilia and flagella, membrane-delimited organelles with nine microtubule doublets surrounding two central microtubules (**figure 4.24**). The doublets slide along each other to bend the cilium or flagellum.
- Despite the fact that eucaryotes and prokaryotes differ in many ways (**table 4.2**), they are quite similar metabolically.

Key Terms

autophagic vacuole 81
axoneme 89
basal body 90
cell cycle 87
cell wall 88
chloroplast 85
chromatin 86
chromosome 86
cilia 89
cisternae 79
cristae 83
cytoplasmic matrix 76
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endosymbiotic theory 85
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polyribosomes 83
polysomes 83
primary lysosomes 81
procaryotic cells 91
proteasome 82
pyrenoid 85
residual body 81
rough or granular ER (RER or GER) 79
secondary lysosomes 81
smooth or agranular ER (SER or AER) 79
stroma 85
thylakoid 85

Questions for Thought and Review

1. Describe the structure and function of every eucaryotic organelle discussed in the chapter.
2. Discuss the statement: "The most obvious difference between eucaryotic and procaryotic cells is in their use of membranes." What general roles do membranes play in eucaryotic cells?
3. Describe how the Golgi apparatus distributes proteins it receives from the ER to different organelles.
4. Briefly discuss how the complex of membranous organelles that de Duve calls the "vacuome" functions as a coordinated whole. What is its function?
5. Describe and contrast the ways in which flagella and cilia propel microorganisms through the water.
6. Outline the major differences between procaryotes and eucaryotes. How are they similar?

Critical Thinking Questions

1. *Giardia lamblia* is an example of eucaryotes that contain nuclei, but no mitochondria. How does the existence of *Giardia* affect the endosymbiosis theory? How do you think *Giardia* obtains its energy? Would your answer change if you learned that *Giardia* is parasitic?
2. Would you expect to find organisms with mitochondria, but without nuclei? Why or why not? Support your answer with literature sources.

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PART II

Microbial Nutrition, Growth, and Control

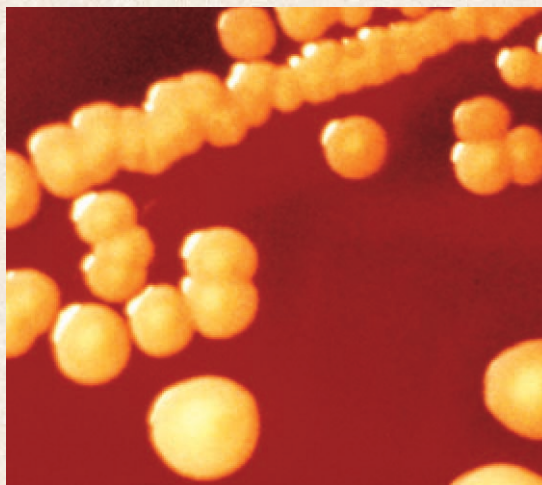
Chapter 5
Microbial Nutrition

Chapter 6
Microbial Growth

Chapter 7
Control of Microorganisms by
Physical and Chemical Agents

CHAPTER 5

Microbial Nutrition



Staphylococcus aureus forms large, golden colonies when growing on blood agar. This human pathogen causes diseases such as boils, abscesses, bacteremia, endocarditis, food poisoning, pharyngitis, and pneumonia.

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Concepts

1. Microorganisms require about 10 elements in large quantities, in part because they are used to construct carbohydrates, lipids, proteins, and nucleic acids. Several other elements are needed in very small amounts and are parts of enzymes and cofactors.
2. All microorganisms can be placed in one of a few nutritional categories on the basis of their requirements for carbon, energy, and hydrogen atoms or electrons.
3. Nutrient molecules frequently cannot cross selectively permeable plasma membranes through passive diffusion. They must be transported by one of three major mechanisms involving the use of membrane carrier proteins. Eucaryotic microorganisms also employ endocytosis for nutrient uptake.
4. Culture media are needed to grow microorganisms in the laboratory and to carry out specialized procedures like microbial identification, water and food analysis, and the isolation of particular microorganisms. Many different media are available for these and other purposes.
5. Pure cultures can be obtained through the use of spread plates, streak plates, or pour plates and are required for the careful study of an individual microbial species.

The whole of nature, as has been said, is a conjugation of the verb to eat, in the active and passive.

—William Ralph Inge

To obtain energy and construct new cellular components, organisms must have a supply of raw materials or nutrients. **Nutrients** are substances used in biosynthesis and energy production and therefore are required for microbial growth. This chapter describes the nutritional requirements of microorganisms, how nutrients are acquired, and the cultivation of microorganisms.

Environmental factors such as temperature, oxygen levels, and the osmotic concentration of the medium are critical in the successful cultivation of microorganisms. These topics are discussed in chapter 6 after an introduction to microbial growth.

5.1 The Common Nutrient Requirements

Analysis of microbial cell composition shows that over 95% of cell dry weight is made up of a few major elements: carbon, oxygen, hydrogen, nitrogen, sulfur, phosphorus, potassium, calcium, magnesium, and iron. These are called **macroelements** or macronutrients because they are required by microorganisms in relatively large amounts. The first six (C, O, H, N, S, and P) are components of carbohydrates, lipids, proteins, and nucleic acids. The remaining four macroelements exist in the cell as cations and play a variety of roles. For example, potassium (K^+) is required for activity by a number of enzymes, including some of those involved in protein synthesis. Calcium (Ca^{2+}), among other functions, contributes to the heat resistance of bacterial endospores. Magnesium (Mg^{2+}) serves as a cofactor for many enzymes, complexes with ATP, and stabilizes ribosomes and cell membranes. Iron (Fe^{2+} and Fe^{3+}) is a part of cytochromes and a cofactor for enzymes and electron-carrying proteins.

All organisms, including microorganisms, require several **micronutrients** or **trace elements** besides macroelements. The micronutrients—manganese, zinc, cobalt, molybdenum, nickel, and copper—are needed by most cells. However, cells require such small amounts that contaminants in water, glassware, and regular media components often are adequate for growth. Therefore it is very difficult to demonstrate a micronutrient requirement. In nature, micronutrients are ubiquitous and probably do not usually limit growth. Micronutrients are normally a part of enzymes and cofactors, and they aid in the catalysis of reactions and maintenance of protein structure. For example, zinc (Zn^{2+}) is present at the active site of some enzymes but is also involved in the association of regulatory and catalytic subunits in *E. coli* aspartate carbamoyltransferase (see section 8.9). Manganese (Mn^{2+}) aids many enzymes catalyzing the transfer of phosphate groups. Molybdenum (Mo^{2+}) is required for nitrogen fixation, and cobalt (Co^{2+}) is a component of vitamin B_{12} . [Electron carriers and enzymes \(pp. 157–64\)](#)

Besides the common macroelements and trace elements, microorganisms may have particular requirements that reflect the special nature of their morphology or environment. Diatoms (see figure 26.6c,d) need silicic acid (H_4SiO_4) to construct their beautiful cell walls of silica [$(SiO_2)_n$]. Although most bacteria do not require large amounts of sodium, many bacteria growing in saline lakes and oceans (see pp. 123, 461) depend on the presence of high concentrations of sodium ion (Na^+).

Finally, it must be emphasized that microorganisms require a balanced mixture of nutrients. If an essential nutrient is in short supply, microbial growth will be limited regardless of the concentrations of other nutrients.

5.2 Requirements for Carbon, Hydrogen, and Oxygen

The requirements for carbon, hydrogen, and oxygen often are satisfied together. Carbon is needed for the skeleton or backbone of all organic molecules, and molecules serving as carbon sources normally also contribute both oxygen and hydrogen atoms. They are the source of all three elements. Because these organic nutrients are almost always reduced and have electrons that they can donate to other molecules, they also can serve as energy sources. Indeed, the more reduced organic molecules are, the higher their energy content (e.g., lipids have a higher energy content than carbohydrates). This is because, as we shall see later, electron transfers release energy when the electrons move from reduced donors with more negative reduction potentials to oxidized electron acceptors with more positive potentials. Thus carbon sources frequently also serve as energy sources, although they don't have to. [Oxidation-reduction reactions and energy \(pp. 157–59\)](#)

One important carbon source that does not supply hydrogen or energy is carbon dioxide (CO_2). This is because CO_2 is oxidized and lacks hydrogen. Probably all microorganisms can fix CO_2 —that is, reduce it and incorporate it into organic molecules. However, by definition, only **autotrophs** can use CO_2 as their sole or principal source of carbon. Many microorganisms are autotrophic, and most of these carry out photosynthesis and use light as their energy source. Some autotrophs oxidize inorganic molecules and derive energy from electron transfers. [Photosynthetic carbon dioxide fixation \(pp. 207–8\)](#)

The reduction of CO_2 is a very energy-expensive process. Thus many microorganisms cannot use CO_2 as their sole carbon source but must rely on the presence of more reduced, complex molecules such as glucose for a supply of carbon. Organisms that use reduced, preformed organic molecules as carbon sources are **heterotrophs** (these preformed molecules normally come from other organisms). As mentioned previously, most heterotrophs use reduced organic compounds as sources of both carbon and energy. For example, the glycolytic pathway produces carbon skeletons for use in biosynthesis and also releases energy as ATP and NADH. [The glycolytic pathway \(pp. 176–77\)](#)

A most remarkable nutritional characteristic of microorganisms is their extraordinary flexibility with respect to carbon sources. Laboratory experiments indicate that there is no natu-

rally occurring organic molecule that cannot be used by some microorganism. Actinomycetes will degrade amyl alcohol, paraffin, and even rubber. Some bacteria seem able to employ almost anything as a carbon source; for example, *Burkholderia cepacia* can use over 100 different carbon compounds. In contrast to these bacterial omnivores, some bacteria are exceedingly fastidious and catabolize only a few carbon compounds. Cultures of methylophilic bacteria metabolize methane, methanol, carbon monoxide, formic acid, and related one-carbon molecules. Parasitic members of the genus *Leptospira* use only long-chain fatty acids as their major source of carbon and energy.

It appears that in natural environments complex populations of microorganisms often will metabolize even relatively indigestible human-made substances such as pesticides. Indigestible molecules sometimes are oxidized and degraded in the presence of a growth-promoting nutrient that is metabolized at the same time, a process called cometabolism. The products of this breakdown process can then be used as nutrients by other microorganisms. [Degradation and microorganisms](#) (pp. 1010–14)

5.3 Nutritional Types of Microorganisms

In addition to the need for carbon, hydrogen, and oxygen, all organisms require sources of energy and electrons for growth to take place. Microorganisms can be grouped into nutritional classes based on how they satisfy all these requirements (**table 5.1**). We have already seen that microorganisms can be classified as either heterotrophs or autotrophs with respect to their preferred source of carbon. There are only two sources of energy available to organisms: (1) light energy, and (2) the energy derived from oxidizing organic or inorganic molecules. **Phototrophs** use light as their energy source; **chemotrophs** obtain energy from the oxidation of chemical compounds (either organic or inorganic). Microorganisms also have only two sources for

electrons. **Lithotrophs** (i.e., “rock-eaters”) use reduced inorganic substances as their electron source, whereas **organotrophs** extract electrons from organic compounds. [Photosynthesis light reactions](#) (pp. 195–201); [Oxidation of organic and inorganic molecules](#) (pp. 176–95)

Despite the great metabolic diversity seen in microorganisms, most may be placed in one of four nutritional classes based on their primary sources of carbon, energy, and electrons (**table 5.2**). The large majority of microorganisms thus far studied are either photolithotrophic autotrophs or chemoorganotrophic heterotrophs. **Photolithotrophic autotrophs** (often called **photoautotrophs** or photolithoautotrophs) use light energy and have CO₂ as their carbon source. Eucaryotic algae and cyanobacteria employ water as the electron donor and release oxygen. Purple and green sulfur

Table 5.1 Sources of Carbon, Energy, and Electrons

Carbon Sources	
Autotrophs	CO ₂ sole or principal biosynthetic carbon source (pp. 207–8) ^a
Heterotrophs	Reduced, preformed, organic molecules from other organisms (chapters 9 and 10)
Energy Sources	
Phototrophs	Light (pp. 195–201)
Chemotrophs	Oxidation of organic or inorganic compounds (chapter 9)
Electron Sources	
Lithotrophs	Reduced inorganic molecules (pp. 193–94)
Organotrophs	Organic molecules (chapter 9)

^aFor each category, the location of material describing the participating metabolic pathways is given within the parentheses.

Table 5.2 Major Nutritional Types of Microorganisms

Major Nutritional Types ^a	Sources of Energy, Hydrogen/Electrons, and Carbon	Representative Microorganisms
Photolithotrophic autotrophy (Photolithoautotrophy)	Light energy Inorganic hydrogen/electron (H/e ⁻) donor CO ₂ carbon source	Algae Purple and green sulfur bacteria Cyanobacteria
Photoorganotrophic heterotrophy (Photoorganoheterotrophy)	Light energy Organic H/e ⁻ donor Organic carbon source (CO ₂ may also be used)	Purple nonsulfur bacteria Green nonsulfur bacteria
Chemolithotrophic autotrophy (Chemolithoautotrophy)	Chemical energy source (inorganic) Inorganic H/e ⁻ donor CO ₂ carbon source	Sulfur-oxidizing bacteria Hydrogen bacteria Nitrifying bacteria Iron-oxidizing bacteria
Chemoorganotrophic heterotrophy (Chemoorganoheterotrophy)	Chemical energy source (organic) Organic H/e ⁻ donor Organic carbon source	Protozoa Fungi Most nonphotosynthetic bacteria (including most pathogens)

^aBacteria in other nutritional categories have been found. The categories are defined in terms of energy, electron, and carbon sources. Condensed versions of these names are given in parentheses.

bacteria cannot oxidize water but extract electrons from inorganic donors like hydrogen, hydrogen sulfide, and elemental sulfur. **Chemoorganotrophic heterotrophs** (often called **chemoheterotrophs**, chemoorganoheterotrophs, or even heterotrophs) use organic compounds as sources of energy, hydrogen, electrons, and carbon. Frequently the same organic nutrient will satisfy all these requirements. It should be noted that essentially all pathogenic microorganisms are chemoheterotrophs.

The other two nutritional classes have fewer microorganisms but often are very important ecologically. Some purple and green bacteria are photosynthetic and use organic matter as their electron donor and carbon source. These **photoorganotrophic heterotrophs** (photoorganoheterotrophs) are common inhabitants of polluted lakes and streams. Some of these bacteria also can grow as photoautotrophs with molecular hydrogen as an electron donor. The fourth group, the **chemolithotrophic autotrophs** (chemolithoautotrophs), oxidizes reduced inorganic compounds such as iron, nitrogen, or sulfur molecules to derive both energy and electrons for biosynthesis. Carbon dioxide is the carbon source. A few chemolithotrophs can derive their carbon from organic sources and thus are heterotrophic. Chemolithotrophs contribute greatly to the chemical transformations of elements (e.g., the conversion of ammonia to nitrate or sulfur to sulfate) that continually occur in the ecosystem. [Photosynthetic and chemolithotrophic bacteria \(sections 21.3, 22.1, and 22.3\)](#)

Although a particular species usually belongs in only one of the four nutritional classes, some show great metabolic flexibility and alter their metabolic patterns in response to environmental changes. For example, many purple nonsulfur bacteria (*see section 22.1*) act as photoorganotrophic heterotrophs in the absence of oxygen but oxidize organic molecules and function chemotrophically at normal oxygen levels. When oxygen is low, photosynthesis and oxidative metabolism may function simultaneously. Another example is provided by bacteria such as *Beggiatoa* (*see p. 501*) that rely on inorganic energy sources and organic (or sometimes CO₂) carbon sources. These microbes are sometimes called **mixotrophic** because they combine chemolithoautotrophic and heterotrophic metabolic processes. This sort of flexibility seems complex and confusing, yet it gives its possessor a definite advantage if environmental conditions frequently change.

1. What are nutrients, and on what basis are they divided into macroelements and micronutrients or trace elements? Describe some ways in which macroelements and micronutrients are used by an organism.
2. Define autotroph and heterotroph.
3. Discuss the ways in which microorganisms are classified based on their requirements for energy and electrons.
4. Describe the nutritional requirements of the four major nutritional groups and give some microbial examples of each. What is a mixotroph?

5.4 Requirements for Nitrogen, Phosphorus, and Sulfur

To grow, a microorganism must be able to incorporate large quantities of nitrogen, phosphorus, and sulfur. Although these elements may be acquired from the same nutrients that supply carbon, microorganisms usually employ inorganic sources as well. [Biochemical mechanisms for the incorporation of nitrogen, phosphorus, and sulfur \(pp. 210–14\)](#)

Nitrogen is needed for the synthesis of amino acids, purines, pyrimidines, some carbohydrates and lipids, enzyme cofactors, and other substances. Many microorganisms can use the nitrogen in amino acids, and ammonia often is directly incorporated through the action of such enzymes as glutamate dehydrogenase or glutamine synthetase and glutamate synthase (*see section 10.4*). Most phototrophs and many nonphotosynthetic microorganisms reduce nitrate to ammonia and incorporate the ammonia in assimilatory nitrate reduction (*see pp. 210–11*). A variety of bacteria (e.g., many cyanobacteria and the symbiotic bacterium *Rhizobium*) can reduce and assimilate atmospheric nitrogen using the nitrogenase system (*see section 10.4*).

Phosphorus is present in nucleic acids, phospholipids, nucleotides like ATP, several cofactors, some proteins, and other cell components. Almost all microorganisms use inorganic phosphate as their phosphorus source and incorporate it directly. Low phosphate levels actually limit microbial growth in many aquatic environments. Phosphate uptake by *E. coli* has been intensively studied. This bacterium can use both organic and inorganic phosphate. Some organophosphates such as hexose 6-phosphates can be taken up directly by transport proteins. Other organophosphates are often hydrolyzed in the periplasm by the enzyme alkaline phosphatase to produce inorganic phosphate, which then is transported across the plasma membrane. When inorganic phosphate is outside the bacterium, it crosses the outer membrane by the use of a porin protein channel. One of two transport systems subsequently moves the phosphate across the plasma membrane. At high phosphate concentrations, transport probably is due to the Pit system. When phosphate concentrations are low, the PST, (phosphate-specific transport) system is more important. The PST system has higher affinity for phosphate; it is an ABC transporter (*see pp. 101–2*) and uses a periplasmic binding protein.

Sulfur is needed for the synthesis of substances like the amino acids cysteine and methionine, some carbohydrates, biotin, and thiamine. Most microorganisms use sulfate as a source of sulfur and reduce it by assimilatory sulfate reduction (*see section 10.4*); a few require a reduced form of sulfur such as cysteine.

5.5 Growth Factors

Microorganisms often grow and reproduce when minerals and sources of energy, carbon, nitrogen, phosphorus, and sulfur are supplied. These organisms have the enzymes and pathways nec-

Table 5.3 Functions of Some Common Vitamins in Microorganisms

Vitamin	Functions	Examples of Microorganisms Requiring Vitamin ^a
Biotin	Carboxylation (CO ₂ fixation) One-carbon metabolism	<i>Leuconostoc mesenteroides</i> (B) <i>Saccharomyces cerevisiae</i> (F) <i>Ochromonas malhamensis</i> (A) <i>Acanthamoeba castellanii</i> (P)
Cyanocobalamin (B ₁₂)	Molecular rearrangements One-carbon metabolism—carries methyl groups	<i>Lactobacillus</i> spp. (B) <i>Euglena gracilis</i> (A) Diatoms and many other algae (A) <i>Acanthamoeba castellanii</i> (P)
Folic acid	One-carbon metabolism	<i>Enterococcus faecalis</i> (B) <i>Tetrahymena pyriformis</i> (P)
Lipoic acid	Transfer of acyl groups	<i>Lactobacillus casei</i> (B) <i>Tetrahymena</i> spp. (P)
Pantothenic acid	Precursor of coenzyme A—carries acyl groups (pyruvate oxidation, fatty acid metabolism)	<i>Proteus morganii</i> (B) <i>Hanseniaspora</i> spp. (F) <i>Paramecium</i> spp. (P)
Pyridoxine (B ₆)	Amino acid metabolism (e.g., transamination)	<i>Lactobacillus</i> spp. (B) <i>Tetrahymena pyriformis</i> (P)
Niacin (nicotinic acid)	Precursor of NAD and NADP—carry electrons and hydrogen atoms	<i>Brucella abortus</i> , <i>Haemophilus influenzae</i> (B) <i>Blastocladiella pringsheimii</i> (F) <i>Crithidia fasciculata</i> (P)
Riboflavin (B ₂)	Precursor of FAD and FMN—carry electrons or hydrogen atoms	<i>Caulobacter vibrioides</i> (B) <i>Dictyostelium</i> spp. (F) <i>Tetrahymena pyriformis</i> (P)
Thiamine (B ₁)	Aldehyde group transfer (pyruvate decarboxylation, α -keto acid oxidation)	<i>Bacillus anthracis</i> (B) <i>Phycomyces blakesleeanus</i> (F) <i>Ochromonas malhamensis</i> (A) <i>Colpidium campylum</i> (P)

^aThe representative microorganisms are members of the following groups: bacteria (B), fungi (F), algae (A), and protozoa (P).

essary to synthesize all cell components required for their well-being. Many microorganisms, on the other hand, lack one or more essential enzymes. Therefore they cannot manufacture all indispensable constituents but must obtain them or their precursors from the environment. Organic compounds required because they are essential cell components or precursors of such components and cannot be synthesized by the organism are called **growth factors**. There are three major classes of growth factors: (1) amino acids, (2) purines and pyrimidines, and (3) vitamins. Amino acids are needed for protein synthesis, purines and pyrimidines for nucleic acid synthesis. **Vitamins** are small organic molecules that usually make up all or part of enzyme cofactors (see section 8.6), and only very small amounts sustain growth. The functions of selected vitamins, and examples of microorganisms requiring them, are given in **table 5.3**. Some microorganisms require many vitamins; for example, *Enterococcus faecalis* needs eight different vitamins for growth. Other growth factors are also seen; heme (from hemoglobin or cytochromes) is required by *Haemophilus influenzae*, and some mycoplasmas need cholesterol.

Knowledge of the specific growth factor requirements of many microorganisms makes possible quantitative growth-response assays for a variety of substances. For example, species from the bacterial genera *Lactobacillus* and *Streptococcus* can be

used in microbiological assays of most vitamins and amino acids. The appropriate bacterium is grown in a series of culture vessels, each containing medium with an excess amount of all required components except the growth factor to be assayed. A different amount of growth factor is added to each vessel. The standard curve is prepared by plotting the growth factor quantity or concentration against the total extent of bacterial growth. Ideally the amount of growth resulting is directly proportional to the quantity of growth factor present; if the growth factor concentration doubles, the final extent of bacterial growth doubles. The quantity of the growth factor in a test sample is determined by comparing the extent of growth caused by the unknown sample with that resulting from the standards. Microbiological assays are specific, sensitive, and simple. They still are used in the assay of substances like vitamin B₁₂ and biotin, despite advances in chemical assay techniques.

The observation that many microorganisms can synthesize large quantities of vitamins has led to their use in industry. Several water-soluble and fat-soluble vitamins are produced partly or completely using industrial fermentations. Good examples of such vitamins and the microorganisms that synthesize them are riboflavin (*Clostridium*, *Candida*, *Ashbya*, *Eremothecium*), coenzyme A (*Brevibacterium*), vitamin B₁₂ (*Streptomyces*, *Propionibacterium*,

Pseudomonas), vitamin C (*Gluconobacter*, *Erwinia*, *Corynebacterium*), β -carotene (*Dunaliella*), and vitamin D (*Saccharomyces*). Current research focuses on improving yields and finding microorganisms that can produce large quantities of other vitamins.

1. Briefly summarize the ways in which microorganisms obtain nitrogen, phosphorus, and sulfur from their environment.
2. What are growth factors? What are vitamins? How can microorganisms be used to determine the quantity of a specific substance in a sample?

5.6 Uptake of Nutrients by the Cell

The first step in nutrient use is uptake of the required nutrients by the microbial cell. Uptake mechanisms must be specific—that is, the necessary substances, and not others, must be acquired. It does a cell no good to take in a substance that it cannot use. Since microorganisms often live in nutrient-poor habitats, they must be able to transport nutrients from dilute solutions into the cell against a concentration gradient. Finally, nutrient molecules must pass through a selectively permeable plasma membrane that will not permit the free passage of most substances. In view of the enormous variety of nutrients and the complexity of the task, it is not surprising that microorganisms make use of several different transport mechanisms. The most important of these are facilitated diffusion, active transport, and group translocation. Eucaryotic microorganisms do not appear to employ group translocation but take up nutrients by the process of endocytosis (see section 4.5). [Plasma membrane structure and properties \(pp. 46–48\)](#)

Facilitated Diffusion

A few substances, such as glycerol, can cross the plasma membrane by **passive diffusion**. Passive diffusion, often simply called diffusion, is the process in which molecules move from a region of higher concentration to one of lower concentration because of random thermal agitation. The rate of passive diffusion is dependent on the size of the concentration gradient between a cell's exterior and its interior (**figure 5.1**). A fairly large concentration gradient is required for adequate nutrient uptake by passive diffusion (i.e., the external nutrient concentration must be high), and the rate of uptake decreases as more nutrient is acquired unless it is used immediately. Very small molecules such as H_2O , O_2 , and CO_2 often move across membranes by passive diffusion. Larger molecules, ions, and polar substances do not cross membranes by passive or simple diffusion.

The rate of diffusion across selectively permeable membranes is greatly increased by using carrier proteins, sometimes called **permeases**, which are embedded in the plasma membrane. Because a carrier aids the diffusion process, it is called **facilitated diffusion**. The rate of facilitated diffusion increases with the concentration gradient much more rapidly and at lower concentrations of the diffusing molecule than that of passive diffusion (fig-

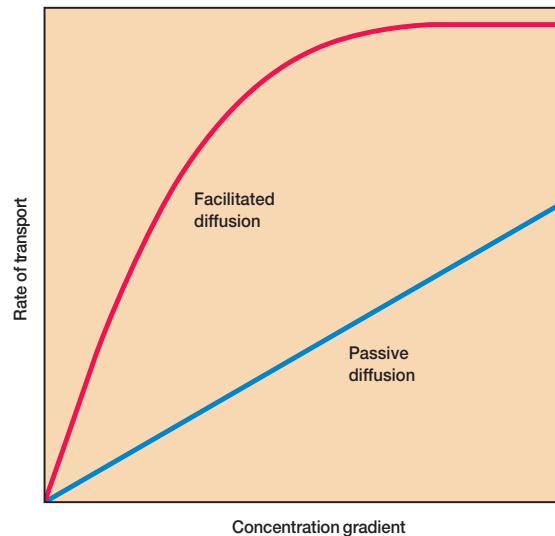


Figure 5.1 **Passive and Facilitated Diffusion.** The dependence of diffusion rate on the size of the solute's concentration gradient. Note the saturation effect or plateau above a specific gradient value when a facilitated diffusion carrier is operating. This saturation effect is seen whenever a carrier protein is involved in transport.

ure 5.1). Note that the diffusion rate levels off or reaches a plateau above a specific gradient value because the carrier is saturated—that is, the carrier protein is binding and transporting as many solute molecules as possible. The resulting curve resembles an enzyme-substrate curve (see section 8.6) and is different from the linear response seen with passive diffusion. Carrier proteins also resemble enzymes in their specificity for the substance to be transported; each carrier is selective and will transport only closely related solutes. Although a carrier protein is involved, facilitated diffusion is truly diffusion. A concentration gradient spanning the membrane drives the movement of molecules, and no metabolic energy input is required. If the concentration gradient disappears, net inward movement ceases. The gradient can be maintained by transforming the transported nutrient to another compound or by moving it to another membranous compartment in eucaryotes. Interestingly, some of these carriers are related to the major intrinsic protein of mammalian eye lenses and thus belong to the MIP family of proteins. The two most widespread MIP channels in bacteria are aquaporins that transport water and glycerol facilitators, which aid glycerol diffusion.

Although much work has been done on the mechanism of facilitated diffusion, the process is not yet understood completely. It appears that the carrier protein complex spans the membrane (**figure 5.2**). After the solute molecule binds to the outside, the carrier may change conformation and release the molecule on the cell interior. The carrier would subsequently change back to its original shape and be ready to pick up another molecule. The net effect is that a lipid-insoluble molecule can enter the cell in response to its concentration gradient. Remember that the mecha-

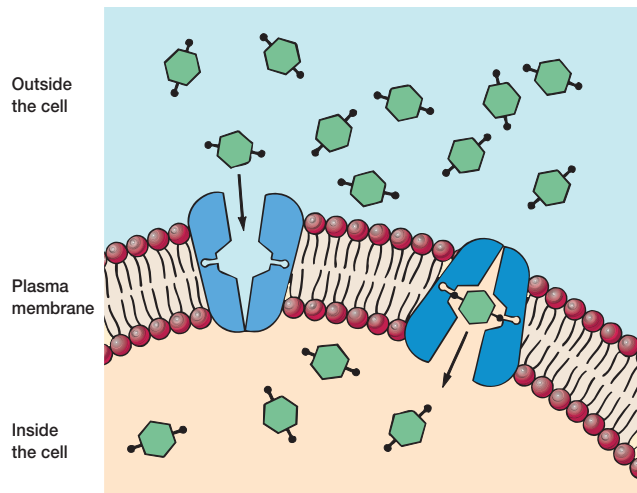


Figure 5.2 A Model of Facilitated Diffusion. The membrane carrier can change conformation after binding an external molecule and subsequently release the molecule on the cell interior. It then returns to the outward oriented position and is ready to bind another solute molecule. Because there is no energy input, molecules will continue to enter only as long as their concentration is greater on the outside.

nism is driven by concentration gradients and therefore is reversible. If the solute's concentration is greater inside the cell, it will move outward. Because the cell metabolizes nutrients upon entry, influx is favored.

Facilitated diffusion does not seem to be important in prokaryotes because nutrient concentrations often are lower outside the cell so that facilitated diffusion cannot be used in uptake. Glycerol is transported by facilitated diffusion in *E. coli*, *Salmonella typhimurium*, *Pseudomonas*, *Bacillus*, and many other bacteria. The process is much more prominent in eucaryotic cells where it is used to transport a variety of sugars and amino acids.

Active Transport

Although facilitated diffusion carriers can efficiently move molecules to the interior when the solute concentration is higher on the outside of the cell, they cannot take up solutes that are already more concentrated within the cell (i.e., against a concentration gradient). Microorganisms often live in habitats characterized by very dilute nutrient sources, and, to flourish, they must be able to transport and concentrate these nutrients. Thus facilitated diffusion mechanisms are not always adequate, and other approaches must be used. The two most important transport processes in such situations are active transport and group translocation, both energy-dependent processes.

Active transport is the transport of solute molecules to higher concentrations, or against a concentration gradient, with the use of metabolic energy input. Because active transport involves protein carrier activity, it resembles facilitated diffusion in

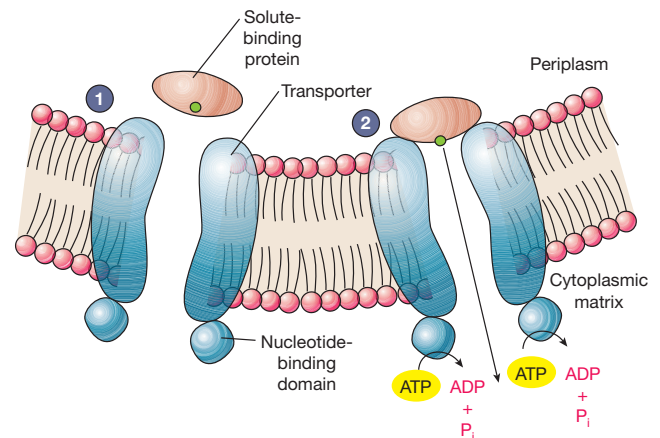


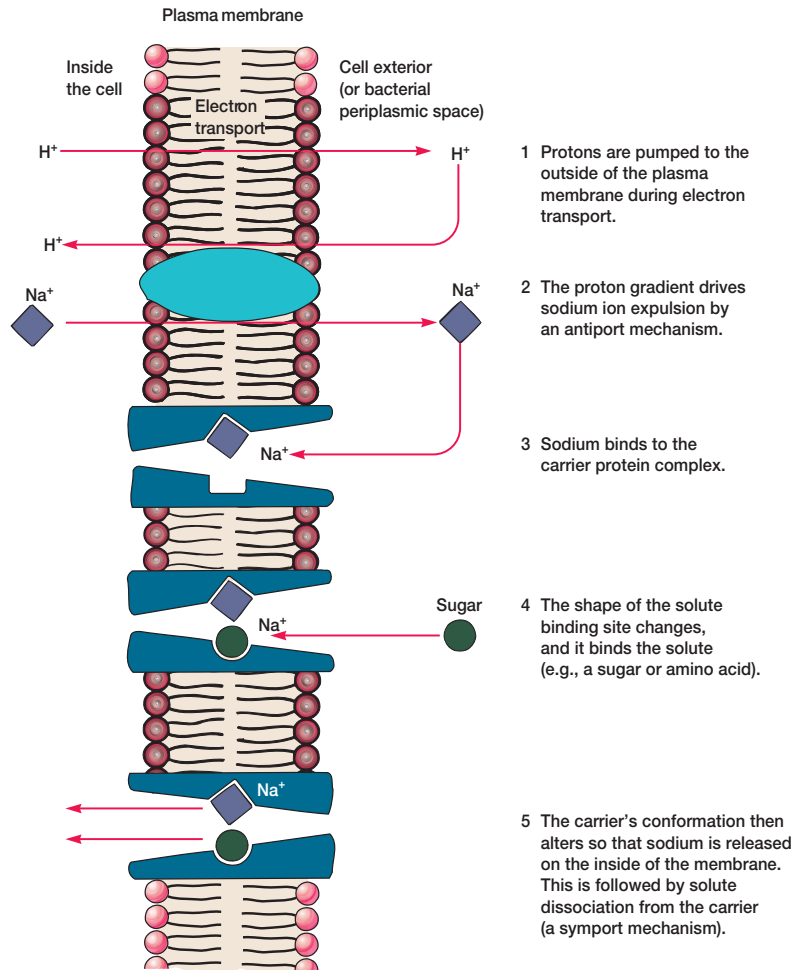
Figure 5.3 ABC Transporter Function. (1) The solute binding protein binds the substrate to be transported and approaches the ABC transporter complex. (2) The solute binding protein attaches to the transporter and releases the substrate, which is moved across the membrane with the aid of ATP hydrolysis. See text for details.

some ways. The carrier proteins or permeases bind particular solutes with great specificity for the molecules transported. Similar solute molecules can compete for the same carrier protein in both facilitated diffusion and active transport. Active transport is also characterized by the carrier saturation effect at high solute concentrations (figure 5.1). Nevertheless, active transport differs from facilitated diffusion in its use of metabolic energy and in its ability to concentrate substances. Metabolic inhibitors that block energy production will inhibit active transport but will not affect facilitated diffusion (at least for a short time).

Binding protein transport systems or **ATP-binding cassette transporters (ABC transporters)** are active in bacteria, archaea, and eucaryotes. Usually these transporters consist of two hydrophobic membrane-spanning domains associated on their cytoplasmic surfaces with two nucleotide-binding domains (**figure 5.3**). The membrane-spanning domains form a pore in the membrane and the nucleotide-binding domains bind and hydrolyze ATP to drive uptake. ABC transporters employ special substrate binding proteins, which are located in the periplasmic space of gram-negative bacteria (see figure 3.23) or are attached to membrane lipids on the external face of the gram-positive plasma membrane. These binding proteins, which also may participate in chemotaxis (see pp. 66–68), bind the molecule to be transported and then interact with the membrane transport proteins to move the solute molecule inside the cell. *E. coli* transports a variety of sugars (arabinose, maltose, galactose, ribose) and amino acids (glutamate, histidine, leucine) by this mechanism.

Substances entering gram-negative bacteria must pass through the outer membrane before ABC transporters and other

Figure 5.4 Active Transport Using Proton and Sodium Gradients. (1) Protons are pumped to the outside of the plasma membrane during electron transport. (2) The proton gradient drives sodium ion expulsion by an antiport mechanism. (3) Sodium binds to the carrier protein complex. (4) The shape of the solute binding site changes, and it binds the solute (e.g., a sugar or amino acid). (5) The carrier's conformation then alters so that sodium is released on the inside of the membrane. This is followed by solute dissociation from the carrier (a symport mechanism).



active transport systems can take action. There are several ways in which this is accomplished. When the substance is small, a generalized porin protein (*see p. 60*) such as OmpF can be used; larger molecules require specialized porins. In some cases (e.g., for uptake of iron and vitamin B₁₂), specialized high-affinity outer membrane receptors and transporters are used.

It should be noted that eucaryotic ABC transporters are sometimes of great medical importance. Some tumor cells pump drugs out using these transporters. Cystic fibrosis results from a mutation that inactivates an ABC transporter that acts as a chloride ion channel in the lungs.

Bacteria also use proton gradients generated during electron transport to drive active transport. The membrane transport proteins responsible for this process lack special periplasmic solute-binding proteins. The lactose permease of *E. coli* is a well-studied example. The permease is a single protein having a molecular weight of about 30,000. It transports a lactose molecule inward as a proton simultaneously enters the cell (a higher concentration of protons is maintained outside the membrane by electron transport chain activity). Such linked transport of two substances in the same direction is called **symport**. Here, energy stored as a proton

gradient drives solute transport. Although the mechanism of transport is not completely understood, it is thought that binding of a proton to the transport protein changes its shape and affinity for the solute to be transported. *E. coli* also uses proton symport to take up amino acids and organic acids like succinate and malate. [The chemiosmotic hypothesis \(p. 187\)](#)

A proton gradient also can power active transport indirectly, often through the formation of a sodium ion gradient. For example, an *E. coli* sodium transport system pumps sodium outward in response to the inward movement of protons (**figure 5.4**). Such linked transport in which the transported substances move in opposite directions is termed **antiport**. The sodium gradient generated by this proton antiport system then drives the uptake of sugars and amino acids. A sodium ion could attach to a carrier protein, causing it to change shape. The carrier would then bind the sugar or amino acid tightly and orient its binding sites toward the cell interior. Because of the low intracellular sodium concentration, the sodium ion would dissociate from the carrier, and the other molecule would follow. *E. coli* transport proteins carry the sugar melibiose and the amino acid glutamate when sodium simultaneously moves inward.

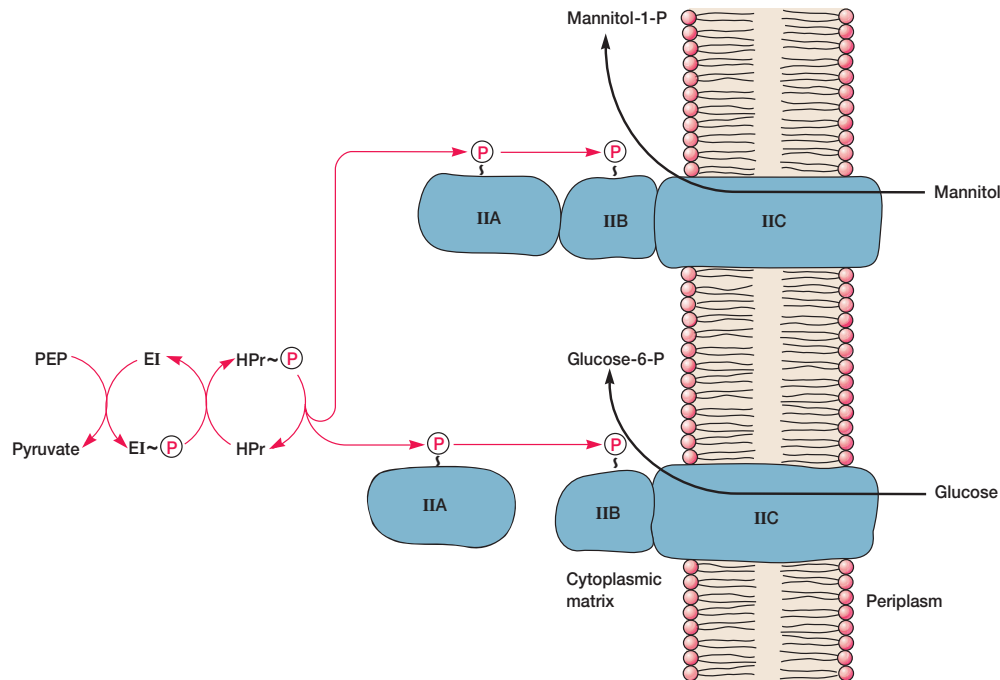


Figure 5.5 Group Translocation: Bacterial PTS Transport. Two examples of the phosphoenolpyruvate: sugar phosphotransferase system (PTS) are illustrated. The following components are involved in the system: phosphoenolpyruvate (PEP), enzyme I (EI), the low molecular weight heat-stable protein (HPr), and enzyme II (EII). The high-energy phosphate is transferred from HPr to the soluble EI_{IA}. EI_{IA} is attached to EI_{IB} in the mannitol transport system and is separate from EI_{IB} in the glucose system. In either case the phosphate moves from EI_{IA} to EI_{IB}, and then is transferred to the sugar during transport through the membrane. Other relationships between the EII components are possible. For example, IIA and IIB may form a soluble protein separate from the membrane complex; the phosphate still moves from IIA to IIB and then to the membrane domain(s).

Sodium symport or cotransport also is an important process in eucaryotic cells where it is used in sugar and amino acid uptake. ATP, rather than proton motive force, usually drives sodium transport in eucaryotic cells.

Often a microorganism has more than one transport system for each nutrient, as can be seen with *E. coli*. This bacterium has at least five transport systems for the sugar galactose, three systems each for the amino acids glutamate and leucine, and two potassium transport complexes. When there are several transport systems for the same substance, the systems differ in such properties as their energy source, their affinity for the solute transported, and the nature of their regulation. Presumably this diversity gives its possessor an added competitive advantage in a variable environment.

Group Translocation

In active transport, solute molecules move across a membrane without modification. Many procaryotes also take up molecules by **group translocation**, a process in which a molecule is transported into the cell while being chemically altered (this can be classified as a type of energy-dependent transport because meta-

bolic energy is used). The best-known group translocation system is the **phosphoenolpyruvate: sugar phosphotransferase system (PTS)**. It transports a variety of sugars into procaryotic cells while phosphorylating them using phosphoenolpyruvate (PEP) as the phosphate donor.



The PTS is quite complex. In *E. coli* and *Salmonella typhimurium*, it consists of two enzymes and a low molecular weight heat-stable protein (HPr). HPr and enzyme I (EI) are cytoplasmic. Enzyme II (EII) is more variable in structure and often composed of three subunits or domains. EI_{IA} (formerly called EI_{II}) is cytoplasmic and soluble. EI_{IB} also is hydrophilic but frequently is attached to EI_{IC}, a hydrophobic protein that is embedded in the membrane. A high-energy phosphate is transferred from PEP to enzyme II with the aid of enzyme I and HPr (figure 5.5). Then, a sugar molecule is phosphorylated as it is carried across the membrane by enzyme II. Enzyme II transports only specific sugars and varies with PTS, whereas enzyme I and HPr are common to all PTSs.

PTSs are widely distributed in procaryotes. Except for some species of *Bacillus* that have both glycolysis and the phosphotransferase system, aerobic bacteria seem to lack PTSs. Members of the

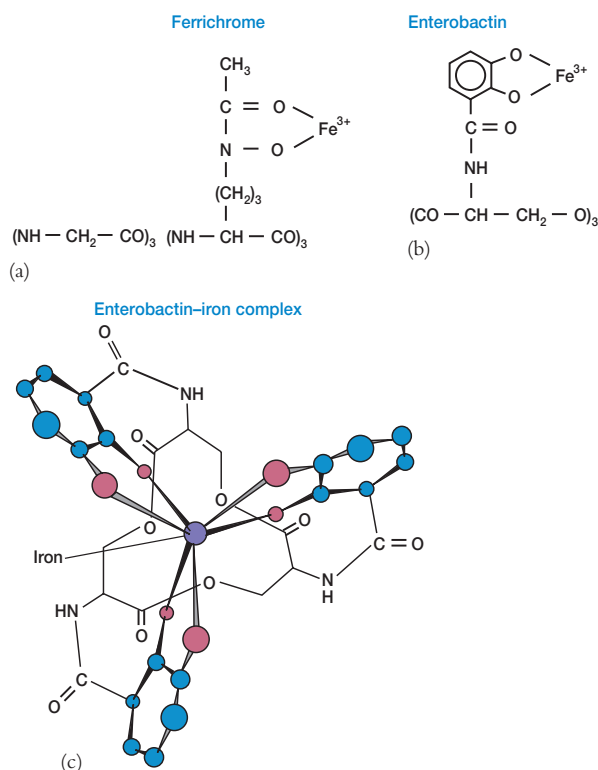


Figure 5.6 Siderophore Ferric Iron Complexes. (a) Ferrichrome is a cyclic hydroxamate [$-\text{CO}-\text{N}(\text{O}^-)-$] molecule formed by many fungi. (b) *E. coli* produces the cyclic catecholate derivative, enterobactin. (c) Ferric iron probably complexes with three siderophore groups to form a six-coordinate, octahedral complex as shown in this illustration of the enterobactin-iron complex.

genera *Escherichia*, *Salmonella*, *Staphylococcus*, and other facultatively anaerobic bacteria (see p. 127) have phosphotransferase systems; some obligately anaerobic bacteria (e.g., *Clostridium*) also have PTSs. Many carbohydrates are transported by these systems. *E. coli* takes up glucose, fructose, mannitol, sucrose, *N*-acetylglucosamine, cellobiose, and other carbohydrates by group translocation. Besides their role in transport, PTS proteins can act as chemoreceptors for chemotaxis.

Iron Uptake

Almost all microorganisms require iron for use in cytochromes and many enzymes. Iron uptake is made difficult by the extreme insolubility of ferric iron (Fe^{3+}) and its derivatives, which leaves little free iron available for transport. Many bacteria and fungi have overcome this difficulty by secreting siderophores [Greek for iron bearers]. **Siderophores** are low molecular weight molecules that are able to complex with ferric iron and supply it to the cell. These iron-transport molecules are normally either hydroxamates or phenolates-catecholates. Ferrichrome is a hydroxamate produced by many fungi; enterobactin is the catecholate formed by *E. coli* (figure 5.6a,b). It

appears that three siderophore groups complex with iron orbitals to form a six-coordinate, octahedral complex (figure 5.6c).

Microorganisms secrete siderophores when little iron is available in the medium. Once the iron-siderophore complex has reached the cell surface, it binds to a siderophore-receptor protein. Then the iron is either released to enter the cell directly or the whole iron-siderophore complex is transported inside by an ABC transporter. In *E. coli* the siderophore receptor is in the outer membrane of the cell envelope; when the iron reaches the periplasmic space, it moves through the plasma membrane with the aid of the transporter. After the iron has entered the cell, it is reduced to the ferrous form (Fe^{2+}). Iron is so crucial to microorganisms that they may use more than one route of iron uptake to ensure an adequate supply.

1. Describe facilitated diffusion, active transport, and group translocation in terms of their distinctive characteristics and mechanisms.
2. How do binding protein transport systems and membrane-bound transport systems differ with respect to energy sources? What are symport and antiport processes?
3. How are siderophores involved in iron transport?

5.7 Culture Media

Much of the study of microbiology depends on the ability to grow and maintain microorganisms in the laboratory, and this is possible only if suitable culture media are available. A culture medium is a solid or liquid preparation used to grow, transport, and store microorganisms. To be effective, the medium must contain all the nutrients the microorganism requires for growth. Specialized media are essential in the isolation and identification of microorganisms, the testing of antibiotic sensitivities, water and food analysis, industrial microbiology, and other activities. Although all microorganisms need sources of energy, carbon, nitrogen, phosphorus, sulfur, and various minerals, the precise composition of a satisfactory medium will depend on the species one is trying to cultivate because nutritional requirements vary so greatly. Knowledge of a microorganism's normal habitat often is useful in selecting an appropriate culture medium because its nutrient requirements reflect its natural surroundings. Frequently a medium is used to select and grow specific microorganisms or to help identify a particular species. In such cases the function of the medium also will determine its composition.

Synthetic or Defined Media

Some microorganisms, particularly photolithotrophic autotrophs such as cyanobacteria and eucaryotic algae, can be grown on relatively simple media containing CO_2 as a carbon source (often added as sodium carbonate or bicarbonate), nitrate or ammonia as a nitrogen source, sulfate, phosphate, and a variety of minerals

Table 5.4 Examples of Defined Media

BG–11 Medium for Cyanobacteria	Amount (g/liter)
NaNO ₃	1.5
K ₂ HPO ₄ ·3H ₂ O	0.04
MgSO ₄ ·7H ₂ O	0.075
CaCl ₂ ·2H ₂ O	0.036
Citric acid	0.006
Ferric ammonium citrate	0.006
EDTA (Na ₂ Mg salt)	0.001
Na ₂ CO ₃	0.02
Trace metal solution ^a	1.0 ml/liter
Final pH 7.4	

Medium for <i>Escherichia coli</i>	Amount (g/liter)
Glucose	1.0
Na ₂ HPO ₄	16.4
KH ₂ PO ₄	1.5
(NH ₄) ₂ SO ₄	2.0
MgSO ₄ ·7H ₂ O	200.0 mg
CaCl ₂	10.0 mg
FeSO ₄ ·7H ₂ O	0.5 mg
Final pH 6.8–7.0	

Sources: Data from Rippka, et al. *Journal of General Microbiology*, 111:1–61, 1979; and S. S. Cohen, and R. Arbogast, *Journal of Experimental Medicine*, 91:619, 1950.

^aThe trace metal solution contains H₃BO₃, MnCl₂·4H₂O, ZnSO₄·7H₂O, Na₂Mo₄·2H₂O, CuSO₄·5H₂O, and Co(NO₃)₂·6H₂O.

(table 5.4). Such a medium in which all components are known is a **defined medium** or **synthetic medium**. Many chemoorganotrophic heterotrophs also can be grown in defined media with glucose as a carbon source and an ammonium salt as a nitrogen source. Not all defined media are as simple as the examples in table 5.4 but may be constructed from dozens of components. Defined media are used widely in research, as it is often desirable to know what the experimental microorganism is metabolizing.

Complex Media

Media that contain some ingredients of unknown chemical composition are **complex media**. Such media are very useful, as a single complex medium may be sufficiently rich and complete to meet the nutritional requirements of many different microorganisms. In addition, complex media often are needed because the nutritional requirements of a particular microorganism are unknown, and thus a defined medium cannot be constructed. This is the situation with many fastidious bacteria, some of which may even require a medium containing blood or serum.

Complex media contain undefined components like peptones, meat extract, and yeast extract. **Peptones** are protein hydrolysates prepared by partial proteolytic digestion of meat, casein, soya meal, gelatin, and other protein sources. They serve as sources of carbon, energy, and nitrogen. Beef extract and yeast extract are aqueous extracts of lean beef and brewer's yeast, respectively. Beef extract contains amino acids, peptides, nucleotides, organic acids, vitamins, and minerals. Yeast extract is an excellent source

Table 5.5 Some Common Complex Media

Nutrient Broth	Amount (g/liter)
Peptone (gelatin hydrolysate)	5
Beef extract	3

Tryptic Soy Broth	
Tryptone (pancreatic digest of casein)	17
Peptone (soybean digest)	3
Glucose	2.5
Sodium chloride	5
Dipotassium phosphate	2.5

MacConkey Agar	
Pancreatic digest of gelatin	17.0
Pancreatic digest of casein	1.5
Peptic digest of animal tissue	1.5
Lactose	10.0
Bile salts	1.5
Sodium chloride	5.0
Neutral red	0.03
Crystal violet	0.001
Agar	13.5

of B vitamins as well as nitrogen and carbon compounds. Three commonly used complex media are (1) nutrient broth, (2) tryptic soy broth, and (3) MacConkey agar (table 5.5).

If a solid medium is needed for surface cultivation of microorganisms, liquid media can be solidified with the addition of 1.0 to 2.0% agar; most commonly 1.5% is used. **Agar** is a sulfated polymer composed mainly of D-galactose, 3,6-anhydro-L-galactose, and D-glucuronic acid (Box 5.1). It usually is extracted from red algae (see figure 26.8). Agar is well suited as a solidifying agent because after it has been melted in boiling water, it can be cooled to about 40 to 42°C before hardening and will not melt again until the temperature rises to about 80 to 90°C. Agar is also an excellent hardening agent because most microorganisms cannot degrade it.

Other solidifying agents are sometimes employed. For example, silica gel is used to grow autotrophic bacteria on solid media in the absence of organic substances and to determine carbon sources for heterotrophic bacteria by supplementing the medium with various organic compounds.

Types of Media

Media such as tryptic soy broth and tryptic soy agar are called general purpose media because they support the growth of many microorganisms. Blood and other special nutrients may be added to general purpose media to encourage the growth of fastidious heterotrophs. These specially fortified media (e.g., blood agar) are called enriched media.

Selective media favor the growth of particular microorganisms. Bile salts or dyes like basic fuchsin and crystal violet favor the growth of gram-negative bacteria by inhibiting the growth of gram-positive bacteria without affecting gram-negative organisms. Endo agar, eosin methylene blue agar, and MacConkey agar

Box 5.1

The Discovery of Agar as a Solidifying Agent and the Isolation of Pure Cultures

The earliest culture media were liquid, which made the isolation of bacteria to prepare pure cultures extremely difficult. In practice, a mixture of bacteria was diluted successively until only one organism, as an average, was present in a culture vessel. If everything went well, the individual bacterium thus isolated would reproduce to give a pure culture. This approach was tedious, gave variable results, and was plagued by contamination problems. Progress in isolating pathogenic bacteria understandably was slow.

The development of techniques for growing microorganisms on solid media and efficiently obtaining pure cultures was due to the efforts of the German bacteriologist Robert Koch and his associates. In 1881 Koch published an article describing the use of boiled potatoes, sliced with a flame-sterilized knife, in culturing bacteria. The surface of a sterile slice of potato was inoculated with bacteria from a needle tip, and then the bacteria were streaked out over the surface so that a few individual cells would be separated from the remainder. The slices were incubated beneath bell jars to prevent airborne contamination, and the isolated cells developed into pure colonies. Unfortunately many bacteria would not grow well on potato slices.

At about the same time, Frederick Loeffler, an associate of Koch's, developed a meat extract peptone medium for cultivating pathogenic

bacteria. Koch decided to try solidifying this medium. Koch was an amateur photographer—he was the first to take photomicrographs of bacteria—and was experienced in preparing his own photographic plates from silver salts and gelatin. Precisely the same approach was employed for preparing solid media. He spread a mixture of Loeffler's medium and gelatin over a glass plate, allowed it to harden, and inoculated the surface in the same way he had inoculated his sliced potatoes. The new solid medium worked well, but it could not be incubated at 37°C (the best temperature for most human bacterial pathogens) because the gelatin would melt. Furthermore, some bacteria digested the gelatin.

About a year later, in 1882, agar was first used as a solidifying agent. It had been discovered by a Japanese innkeeper, Minora Tarazemon. The story goes that he threw out extra seaweed soup and discovered the next day that it had jelled during the cold winter night. Agar had been used by the East Indies Dutch to make jellies and jams. Fannie Eilshemius Hesse (see figure 1.5), the New Jersey-born wife of Walther Hesse, one of Koch's assistants, had learned of agar from a Dutch acquaintance and suggested its use when she heard of the difficulties with gelatin. Agar-solidified medium was an instant success and continues to be essential in all areas of microbiology.

(table 5.5), three media widely used for the detection of *E. coli* and related bacteria in water supplies and elsewhere, contain dyes that suppress gram-positive bacterial growth. MacConkey agar also contains bile salts. Bacteria also may be selected by incubation with nutrients that they specifically can use. A medium containing only cellulose as a carbon and energy source is quite effective in the isolation of cellulose-digesting bacteria. The possibilities for selection are endless, and there are dozens of special selective media in use.

Differential media are media that distinguish between different groups of bacteria and even permit tentative identification of microorganisms based on their biological characteristics. Blood agar is both a differential medium and an enriched one. It distinguishes between hemolytic and nonhemolytic bacteria. Hemolytic bacteria (e.g., many streptococci and staphylococci isolated from throats) produce clear zones around their colonies because of red blood cell destruction. MacConkey agar is both differential and selective. Since it contains lactose and neutral red dye, lactose-fermenting colonies appear pink to red in color and are easily distinguished from colonies of nonfermenters.

1. Describe the following kinds of media and their uses: defined or synthetic media, complex media, general purpose media, enriched media, selective media, and differential media. Give an example of each kind.
2. What are peptones, yeast extract, beef extract, and agar? Why are they used in media?

5.8 Isolation of Pure Cultures

In natural habitats microorganisms usually grow in complex, mixed populations containing several species. This presents a problem for the microbiologist because a single type of microorganism cannot be studied adequately in a mixed culture. One needs a **pure culture**, a population of cells arising from a single cell, to characterize an individual species. Pure cultures are so important that the development of pure culture techniques by the German bacteriologist Robert Koch transformed microbiology. Within about 20 years after the development of pure culture techniques most pathogens responsible for the major human bacterial diseases had been isolated (see Table 1.1). There are several ways to prepare pure cultures; a few of the more common approaches are reviewed here. [A brief survey of some major milestones in microbiology \(chapter 1\)](#)

The Spread Plate and Streak Plate

If a mixture of cells is spread out on an agar surface so that every cell grows into a completely separate **colony**, a macroscopically visible growth or cluster of microorganisms on a solid medium, each colony represents a pure culture. The **spread plate** is an easy, direct way of achieving this result. A small volume of dilute microbial mixture containing around 30 to 300 cells is transferred to the center of an agar plate and spread evenly over the surface with a sterile bent-glass rod (figure 5.7). The dispersed cells develop into isolated colonies. Because the number of colonies

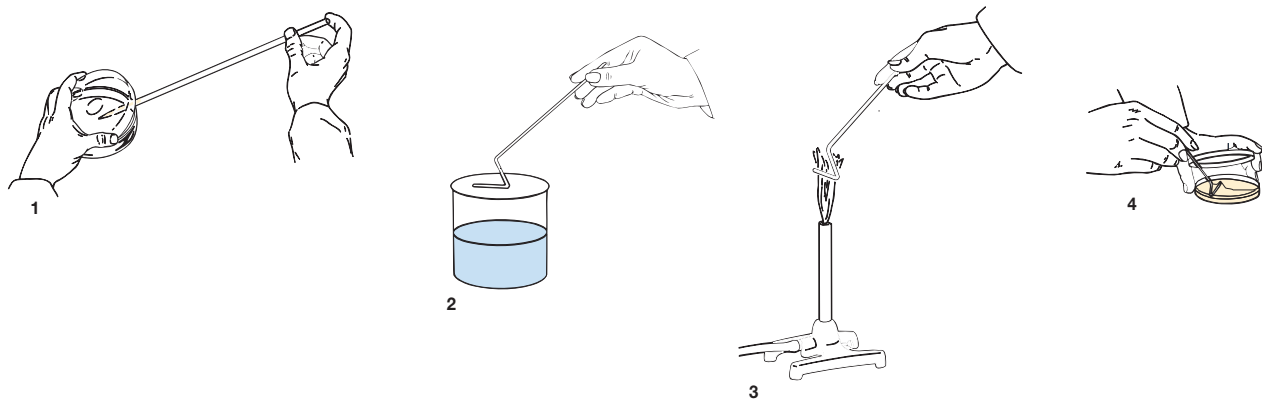


Figure 5.7 Spread-Plate Technique. The preparation of a spread plate. (1) Pipette a small sample onto the center of an agar medium plate. (2) Dip a glass spreader into a beaker of ethanol. (3) Briefly flame the ethanol-soaked spreader and allow it to cool. (4) Spread the sample evenly over the agar surface with the sterilized spreader. Incubate.

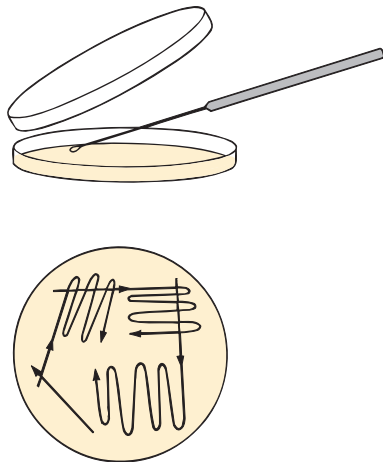


Figure 5.8 Streak-Plate Technique. Preparation of streak plates. The upper illustration shows a petri dish of agar being streaked with an inoculating loop. A commonly used streaking pattern is pictured at the bottom.

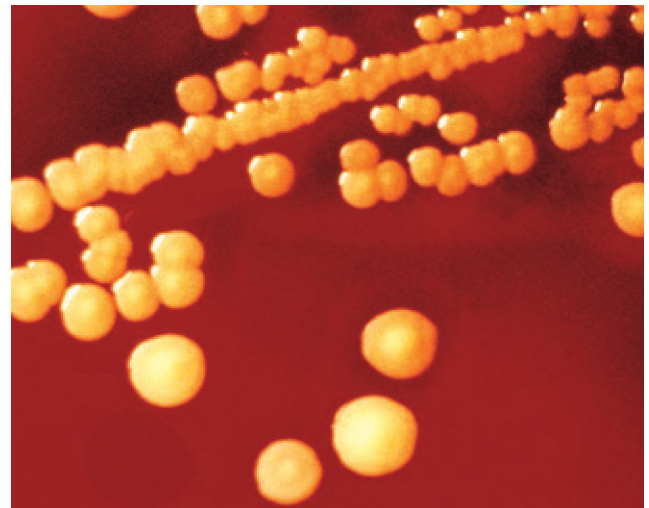


Figure 5.9 Bacterial Colonies on Agar. Colonies growing on a streak plate. A blood-agar plate has been inoculated with *Staphylococcus aureus*. After incubation, large, golden colonies have formed on the agar.

should equal the number of viable organisms in the sample, spread plates can be used to count the microbial population.

Pure colonies also can be obtained from **streak plates**. The microbial mixture is transferred to the edge of an agar plate with an inoculating loop or swab and then streaked out over the surface in one of several patterns (**figure 5.8**). At some point in the process, single cells drop from the loop as it is rubbed along the agar surface and develop into separate colonies (**figure 5.9**). In both spread-plate and streak-plate techniques, successful isolation depends on spatial separation of single cells.

The Pour Plate

Extensively used with bacteria and fungi, a **pour plate** also can yield isolated colonies. The original sample is diluted several times to reduce the microbial population sufficiently to obtain separate colonies when plating (**figure 5.10**). Then small volumes of several diluted samples are mixed with liquid agar that has been cooled to about 45°C, and the mixtures are poured immediately into sterile culture dishes. Most bacteria and fungi are not killed by a brief exposure to the warm agar. After the agar has hardened, each cell is fixed in place and forms an individual

Box 5.2

The Enrichment and Isolation of Pure Cultures

A major practical problem is the preparation of pure cultures when microorganisms are present in very low numbers in a sample. Plating methods can be combined with the use of selective or differential media to enrich and isolate rare microorganisms. A good example is the isolation of bacteria that degrade the herbicide 2,4-dichlorophenoxyacetic acid (2,4-D). Bacteria able to metabolize 2,4-D can be obtained with a liquid medium containing 2,4-D as its sole carbon source and the required nitrogen, phosphorus, sulfur, and mineral components. When this medium is inoculated with soil, only bac-

teria able to use 2,4-D will grow. After incubation, a sample of the original culture is transferred to a fresh flask of selective medium for further enrichment of 2,4-D metabolizing bacteria. A mixed population of 2,4-D degrading bacteria will arise after several such transfers. Pure cultures can be obtained by plating this mixture on agar containing 2,4-D as the sole carbon source. Only bacteria able to grow on 2,4-D form visible colonies and can be subcultured. This same general approach is used to isolate and purify a variety of bacteria by selecting for specific physiological characteristics.

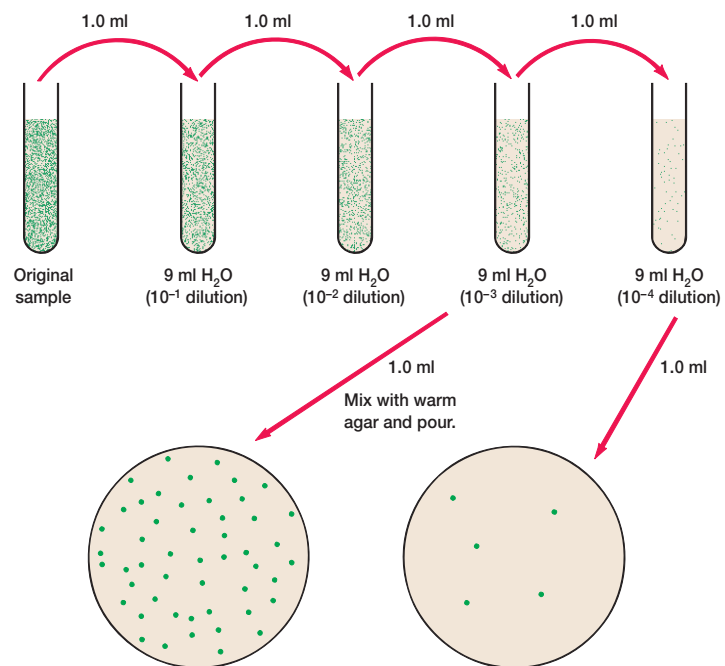


Figure 5.10 The Pour-Plate Technique. The original sample is diluted several times to thin out the population sufficiently. The most diluted samples are then mixed with warm agar and poured into petri dishes. Isolated cells grow into colonies and can be used to establish pure cultures. The surface colonies are circular; subsurface colonies would be lenticular or lens shaped.

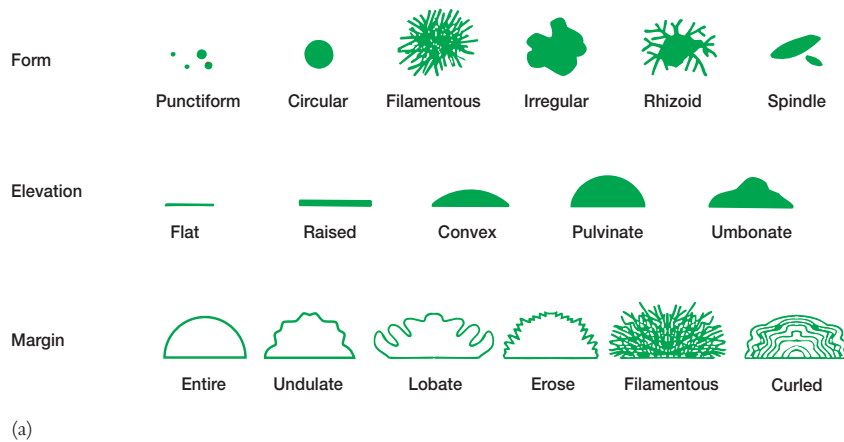
colony. Plates containing between 30 and 300 colonies are counted. The total number of colonies equals the number of viable microorganisms in the diluted sample. Colonies growing on the surface also can be used to inoculate fresh medium and prepare pure cultures (Box 5.2).

The preceding techniques require the use of special culture dishes named **petri dishes** or plates after their inventor Julius Richard Petri, a member of Robert Koch's laboratory; Petri developed these dishes around 1887 and they immediately replaced agar-coated glass plates. They consist of two round halves, the top half overlapping the bottom (figure 5.8). Petri dishes are very easy to use, may be stacked on each other to save space, and are one of the most common items in microbiology laboratories.

Colony Morphology and Growth

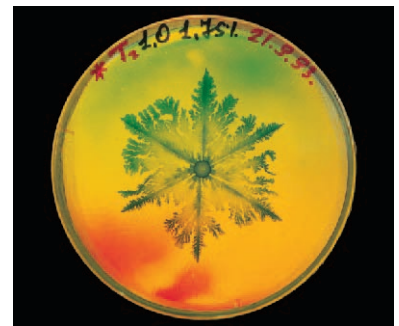
Colony development on agar surfaces aids the microbiologist in identifying bacteria because individual species often form colonies of characteristic size and appearance (figure 5.11). When a mixed population has been plated properly, it sometimes is possible to identify the desired colony based on its overall appearance and use it to obtain a pure culture. The structure of bacterial colonies also has been examined with the scanning electron microscope. The microscopic structure of colonies is often as variable as their visible appearance (figure 5.12).

In nature bacteria and many other microorganisms often grow on surfaces in biofilms. However, sometimes they do form discrete colonies. Therefore an understanding of colony growth is impor-



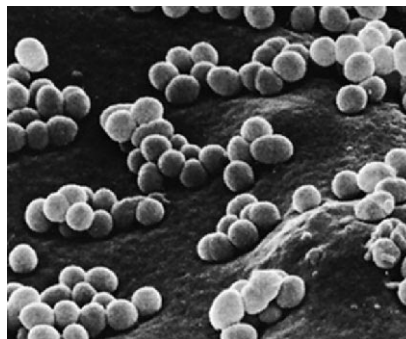
(a)

Figure 5.11 Bacterial Colony Morphology. (a) Variations in bacterial colony morphology seen with the naked eye. The general form of the colony and the shape of the edge or margin can be determined by looking down at the top of the colony. The nature of colony elevation is apparent when viewed from the side as the plate is held at eye level. (b) Colony morphology can vary dramatically with the medium on which the bacteria are growing. These beautiful snowflakelike colonies were formed by *Bacillus subtilis* growing on nutrient-poor agar. The bacteria apparently behave cooperatively when confronted with poor growth conditions, and often the result is an intricate structure that resembles the fractal patterns seen in nonliving systems.



(b)

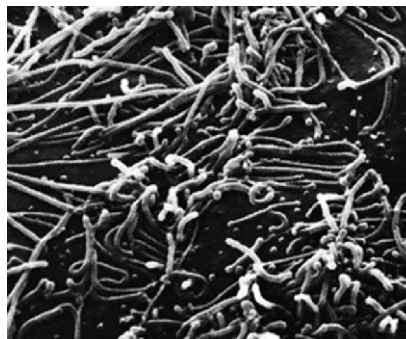
Figure 5.12 Scanning Electron Micrographs of Bacterial Colonies. (a) *Micrococcus* on agar ($\times 31,000$). (b) *Clostridium* ($\times 12,000$). (c) *Mycoplasma pneumoniae* ($\times 26,000$). (d) *Escherichia coli* ($\times 14,000$).



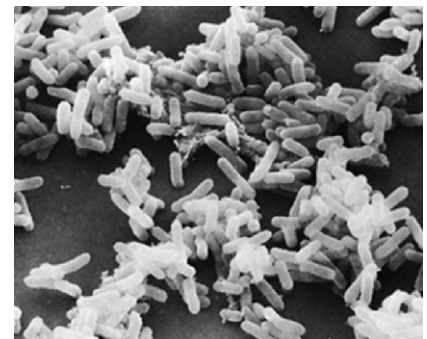
(a)



(b)



(c)



(d)

tant, and the growth of colonies on agar has been frequently studied. Generally the most rapid cell growth occurs at the colony edge. Growth is much slower in the center, and cell autolysis takes place in the older central portions of some colonies. These differences in growth appear due to gradients of oxygen, nutrients, and toxic products within the colony. At the colony edge, oxygen and nutrients are plentiful. The colony center, of course, is much thicker than the edge. Consequently oxygen and nutrients do not diffuse readily into the center, toxic metabolic products cannot be quickly eliminated, and growth in the colony center is slowed or stopped. Because of these environmental variations within a colony, cells on the periphery can be growing at maximum rates while cells in the center are dying. [Biofilms \(pp. 620–22\)](#)

It is obvious from the colonies pictured in figure 5.11 that bacteria growing on solid surfaces such as agar can form quite

complex and intricate colony shapes. These patterns vary with nutrient availability and the hardness of the agar surface. It is not yet clear how characteristic colony patterns develop. Nutrient diffusion and availability, bacterial chemotaxis, and the presence of liquid on the surface all appear to play a role in pattern formation. Undoubtedly cell-cell communication and quorum sensing (*see pp. 132–33*) is important as well. Much work will be required to understand the formation of bacterial colonies and biofilms.

1. What are pure cultures, and why are they important? How are spread plates, streak plates, and pour plates prepared?
2. In what way does microbial growth vary within a colony? What factors might cause these variations in growth?

Summary

1. Microorganisms require nutrients, materials that are used in biosynthesis and energy production.
2. Macronutrients or macroelements (C, O, H, N, S, P, K, Ca, Mg, and Fe) are needed in relatively large quantities; micronutrients or trace elements (e.g., Mn, Zn, Co, Mo, Ni, and Cu) are used in very small amounts.
3. Autotrophs use CO₂ as their primary or sole carbon source; heterotrophs employ organic molecules.
4. Microorganisms can be classified based on their energy and electron sources ([table 5.1](#)). Phototrophs use light energy, and chemotrophs obtain energy from the oxidation of chemical compounds. Electrons are extracted from reduced inorganic substances by lithotrophs and from organic compounds by organotrophs ([table 5.2](#)).
5. Nitrogen, phosphorus, and sulfur may be obtained from the same organic molecules that supply carbon, from the direct incorporation of ammonia and phosphate, and by the reduction and assimilation of oxidized inorganic molecules.
6. Probably most microorganisms need growth factors. Growth factor requirements make microbiological assays possible.
7. Although some nutrients can enter cells by passive diffusion, a membrane carrier protein is usually required.
8. In facilitated diffusion the transport protein simply carries a molecule across the membrane in the direction of decreasing concentration, and no metabolic energy is required ([figure 5.2](#)).
9. Active transport systems use metabolic energy and membrane carrier proteins to concentrate substances actively by transporting them against a gradient. ATP is used as an energy source by ABC transporters ([figure 5.3](#)). Gradients of protons and sodium ions also drive solute uptake across membranes ([figure 5.4](#)).
10. Bacteria also transport organic molecules while modifying them, a process known as group translocation. For example, many sugars are transported and phosphorylated simultaneously ([figure 5.5](#)).
11. Iron is accumulated by the secretion of siderophores, small molecules able to complex with ferric iron ([figure 5.6](#)). When the iron-siderophore complex reaches the cell surface, it is taken inside and the iron is reduced to the ferrous form.
12. Culture media can be constructed completely from chemically defined components (defined media or synthetic media) or may contain constituents like peptones and yeast extract whose precise composition is unknown (complex media).
13. Culture media can be solidified by the addition of agar, a complex polysaccharide from red algae.
14. Culture media are classified based on function and composition as general purpose media, enriched media, selective media, and differential media.
15. Pure cultures usually are obtained by isolating individual cells with any of three plating techniques: the spread-plate, streak-plate, and pour-plate methods ([figures 5.7 and 5.8](#)).
16. Microorganisms growing on solid surfaces tend to form colonies with distinctive morphology. Colonies usually grow most rapidly at the edge where larger amounts of required resources are available.

Key Terms

active transport 101

agar 105

antiport 102

ATP-binding cassette transporters (ABC transporters) 101

autotrophs 96

chemoheterotrophs 98

chemolithotrophic autotrophs 98

chemoorganotrophic heterotrophs 98

chemotrophs 97

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micronutrients 96

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Questions for Thought and Review

1. Why is it so difficult to demonstrate the micronutrient requirements of microorganisms?
2. List some of the most important uses of the nitrogen, phosphorus, and sulfur that microorganisms obtain from their surroundings.
3. Why are amino acids, purines, and pyrimidines often growth factors, whereas glucose is usually not?
4. Why do microorganisms normally take up nutrients using transport proteins or permeases? What advantage does a microorganism gain by employing active transport rather than facilitated diffusion?
5. If you wished to obtain a pure culture of bacteria that could degrade benzene and use it as a carbon and energy source, how would you proceed?
6. Describe the nutritional requirements of a chemolithotrophic heterotroph. Where might you search for such a bacterium?
7. Suppose that you carry out a serial dilution of a 0.1 ml sample as shown in figure 5.10. The 10^{-3} plate gives 80 colonies and the 10^{-4} plate yields four colonies. Calculate the concentration (bacteria/ml) of the original, undiluted sample.

Critical Thinking Questions

1. Discuss the advantages and disadvantages of group translocation versus endocytosis for the host cell.
2. Explain why isolation of a pure culture on selective solid medium may not be successful.

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5.7 Culture Media

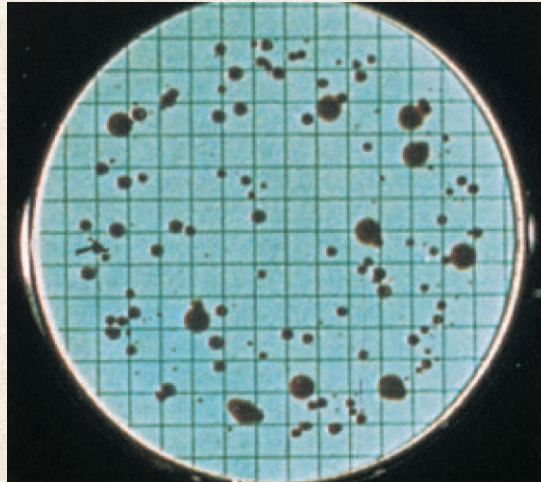
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CHAPTER 6

Microbial Growth



Membrane filters are used in counting microorganisms. This membrane has been used to obtain a total bacterial count using an indicator to color colonies for easy counting.

Outline

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Concepts

1. Growth is defined as an increase in cellular constituents and may result in an increase in a microorganism's size, population number, or both.
2. When microorganisms are grown in a closed system, population growth remains exponential for only a few generations and then enters a stationary phase due to factors such as nutrient limitation and waste accumulation. In an open system with continual nutrient addition and waste removal, the exponential phase can be maintained for long periods.
3. A wide variety of techniques can be used to study microbial growth by following changes in the total cell number, the population of viable microorganisms, or the cell mass.
4. Water availability, pH, temperature, oxygen concentration, pressure, radiation, and a number of other environmental factors influence microbial growth. Yet many microorganisms, and particularly bacteria, have managed to adapt and flourish under environmental extremes that would destroy most higher organisms.
5. In the natural environment, growth is often severely limited by available nutrient supplies and many other environmental factors.
6. Bacteria can communicate with each other and behave cooperatively using population density-dependent signals.

The paramount evolutionary accomplishment of bacteria as a group is rapid, efficient cell growth in many environments.

—J. L. Ingraham, O. Maaløe,
and F. C. Neidhardt

Chapter 5 emphasizes that microorganisms need access to a source of energy and the raw materials essential for the construction of cellular components. All organisms must have carbon, hydrogen, oxygen, nitrogen, sulfur, phosphorus, and a variety of minerals; many also require one or more special growth factors. The cell takes up these substances by membrane transport processes, the most important of which are facilitated diffusion, active transport, and group translocation. Eucaryotic cells also employ endocytosis.

Chapter 6 concentrates more directly on the growth. The nature of growth and the ways in which it can be measured are described first, followed by consideration of continuous culture techniques. An account of the influence of environmental factors on microbial growth completes the chapter.

Growth may be defined as an increase in cellular constituents. It leads to a rise in cell number when microorganisms reproduce by processes like budding or binary fission. In the latter, individual cells enlarge and divide to yield two progeny of approximately equal size. Growth also results when cells simply become longer or larger. If the microorganism is **coenocytic**—that is, a multinucleate organism in which nuclear divisions are not accompanied by cell divisions—growth results in an increase in cell size but not cell number. It is usually not convenient to investigate the growth and reproduction of individual microorganisms because of their small size. Therefore, when studying growth, microbiologists normally follow changes in the total population number. [The cell cycle \(pp. 87; 285–86\)](#)

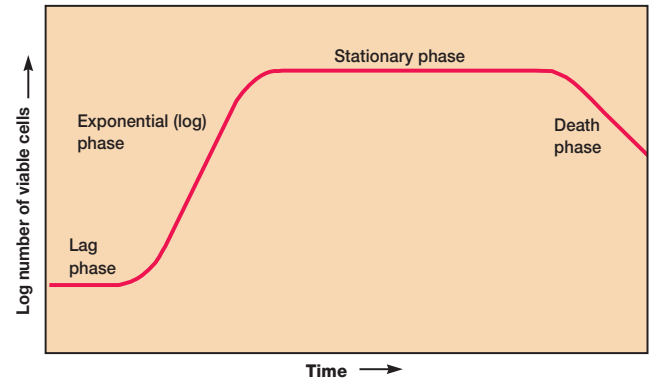


Figure 6.1 Microbial Growth Curve in a Closed System. The four phases of the growth curve are identified on the curve and discussed in the text.

6.1 The Growth Curve

Population growth is studied by analyzing the growth curve of a microbial culture. When microorganisms are cultivated in liquid medium, they usually are grown in a **batch culture** or closed system—that is, they are incubated in a closed culture vessel with a single batch of medium. Because no fresh medium is provided during incubation, nutrient concentrations decline and concentrations of wastes increase. The growth of microorganisms reproducing by binary fission can be plotted as the logarithm of the number of viable cells versus the incubation time. The resulting curve has four distinct phases ([figure 6.1](#)).

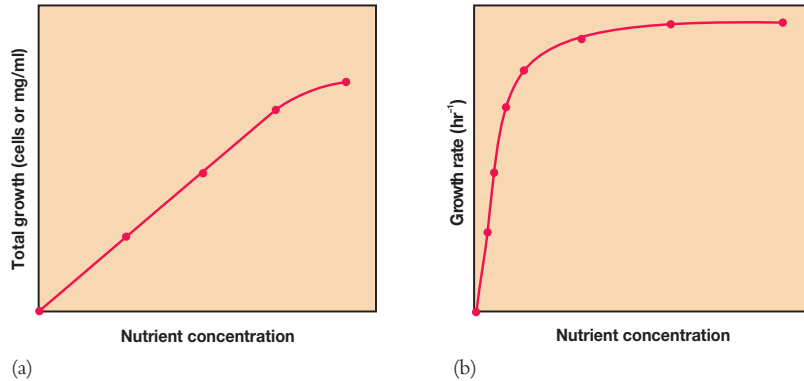
Lag Phase

When microorganisms are introduced into fresh culture medium, usually no immediate increase in cell number occurs, and therefore this period is called the **lag phase**. Although cell division does not take place right away and there is no net increase in mass, the cell is synthesizing new components. A lag phase prior to the start of cell division can be necessary for a variety of reasons. The cells may be old and depleted of ATP, essential cofactors, and ribosomes; these must be synthesized before growth can begin. The medium may be different from the one the microorganism was growing in previously. Here new enzymes would be needed to use different nutrients. Possibly the microorganisms have been injured and require time to recover. Whatever the causes, eventually the cells retool, replicate their DNA, begin to increase in mass, and finally divide.

The lag phase varies considerably in length with the condition of the microorganisms and the nature of the medium. This phase may be quite long if the inoculum is from an old culture or one that has been refrigerated. Inoculation of a culture into a chemically different medium also results in a longer lag phase.

Figure 6.2 Nutrient Concentration and Growth.

(a) The effect of changes in limiting nutrient concentration on total microbial yield. At sufficiently high concentrations, total growth will plateau.
(b) The effect on growth rate.



On the other hand, when a young, vigorously growing exponential phase culture is transferred to fresh medium of the same composition, the lag phase will be short or absent.

Exponential Phase

During the **exponential** or **log phase**, microorganisms are growing and dividing at the maximal rate possible given their genetic potential, the nature of the medium, and the conditions under which they are growing. Their rate of growth is constant during the exponential phase; that is, the microorganisms are dividing and doubling in number at regular intervals. Because each individual divides at a slightly different moment, the growth curve rises smoothly rather than in discrete jumps (figure 6.1). The population is most uniform in terms of chemical and physiological properties during this phase; therefore exponential phase cultures are usually used in biochemical and physiological studies.

Exponential growth is **balanced growth**. That is, all cellular constituents are manufactured at constant rates relative to each other. If nutrient levels or other environmental conditions change, **unbalanced growth** results. This is growth during which the rates of synthesis of cell components vary relative to one another until a new balanced state is reached. This response is readily observed in a shift-up experiment in which bacteria are transferred from a nutritionally poor medium to a richer one. The cells first construct new ribosomes to enhance their capacity for protein synthesis. This is followed by increases in protein and DNA synthesis. Finally, the expected rise in reproductive rate takes place.

[Protein and DNA synthesis \(sections 11.3 and 12.2\)](#)

Unbalanced growth also results when a bacterial population is shifted down from a rich medium to a poor one. The organisms may previously have been able to obtain many cell components directly from the medium. When shifted to a nutritionally inadequate medium, they need time to make the enzymes required for the biosynthesis of unavailable nutrients. Consequently cell division and DNA replication continue after the shift-down, but net protein and RNA synthesis slow. The cells become smaller and reorganize themselves metabolically until they are able to grow again. Then balanced growth is resumed and the culture enters the exponential phase. [Regulation of nucleic acid synthesis \(pp. 275–83\)](#)

These shift-up and shift-down experiments demonstrate that microbial growth is under precise, coordinated control and responds quickly to changes in environmental conditions.

When microbial growth is limited by the low concentration of a required nutrient, the final net growth or yield of cells increases with the initial amount of the limiting nutrient present (**figure 6.2a**). This is the basis of microbiological assays for vitamins and other growth factors. The rate of growth also increases with nutrient concentration (figure 6.2b), but in a hyperbolic manner much like that seen with many enzymes (*see figure 8.17*). The shape of the curve seems to reflect the rate of nutrient uptake by microbial transport proteins. At sufficiently high nutrient levels the transport systems are saturated, and the growth rate does not rise further with increasing nutrient concentration. [Microbiological assays \(p. 99\); Nutrient transport systems \(pp. 100–4\)](#)

Stationary Phase

Eventually population growth ceases and the growth curve becomes horizontal (figure 6.1). This **stationary phase** usually is attained by bacteria at a population level of around 10^9 cells per ml. Other microorganisms normally do not reach such high population densities, protozoan and algal cultures often having maximum concentrations of about 10^6 cells per ml. Of course final population size depends on nutrient availability and other factors, as well as the type of microorganism being cultured. In the stationary phase the total number of viable microorganisms remains constant. This may result from a balance between cell division and cell death, or the population may simply cease to divide though remaining metabolically active.

Microbial populations enter the stationary phase for several reasons. One obvious factor is nutrient limitation; if an essential nutrient is severely depleted, population growth will slow. Aerobic organisms often are limited by O_2 availability. Oxygen is not very soluble and may be depleted so quickly that only the surface of a culture will have an O_2 concentration adequate for growth. The cells beneath the surface will not be able to grow unless the culture is shaken or aerated in another way. Population growth also may cease due to the accumulation of toxic waste products. This factor seems to limit the growth of many anaerobic cultures (cultures growing in the absence of O_2). For example, streptococci can produce so much lactic acid and other organic acids from sugar fermentation that their medium becomes acidic and

growth is inhibited. Streptococcal cultures also can enter the stationary phase due to depletion of their sugar supply. Finally, there is some evidence that growth may cease when a critical population level is reached. Thus entrance into the stationary phase may result from several factors operating in concert.

As we have seen, bacteria in a batch culture may enter stationary phase in response to starvation. This probably often occurs in nature as well because many environments have quite low nutrient levels. Starvation can be a positive experience for bacteria. Many do not respond with obvious morphological changes such as endospore formation, but only decrease somewhat in overall size, often accompanied by protoplast shrinkage and nucleoid condensation. The more important changes are in gene expression and physiology. Starving bacteria frequently produce a variety of **starvation proteins**, which make the cell much more resistant to damage in a variety of ways. They increase peptidoglycan cross-linking and cell wall strength. The Dps (*DNA-binding protein from starved cells*) protein protects DNA. Chaperones prevent protein denaturation and renature damaged proteins. As a result of these and many other mechanisms, the starved cells become harder to kill and more resistant to starvation itself, damaging temperature changes, oxidative and osmotic damage, and toxic chemicals such as chlorine. These changes are so effective that some bacteria can survive starvation for years. Clearly, these considerations are of great practical importance in medical and industrial microbiology. There is even evidence that *Salmonella typhimurium* and some other bacterial pathogens become more virulent when starved.

Death Phase

Detrimental environmental changes like nutrient deprivation and the buildup of toxic wastes lead to the decline in the number of viable cells characteristic of the **death phase**. The death of a microbial population, like its growth during the exponential phase, is usually logarithmic (that is, a constant proportion of cells dies every hour). This pattern in viable cell count holds even when the total cell number remains constant because the cells simply fail to lyse after dying. Often the only way of deciding whether a bacterial cell is viable is by incubating it in fresh medium; if it does not grow and reproduce, it is assumed to be dead. That is, death is defined to be the irreversible loss of the ability to reproduce.

Although most of a microbial population usually dies in a logarithmic fashion, the death rate may decrease after the population has been drastically reduced. This is due to the extended survival of particularly resistant cells. For this and other reasons, the death phase curve may be complex.

The Mathematics of Growth

Knowledge of microbial growth rates during the exponential phase is indispensable to microbiologists. Growth rate studies contribute to basic physiological and ecological research and the solution of applied problems in industry. Therefore the quantitative aspects of exponential phase growth will be discussed.

During the exponential phase each microorganism is dividing at constant intervals. Thus the population will double in number during a specific length of time called the **generation time** or **dou-**

Table 6.1 An Example of Exponential Growth

Time ^a	Division Number	2 ⁿ	Population (N ₀ × 2 ⁿ)	log ₁₀ N _t
0	0	2 ⁰ = 1	1	0.000
20	1	2 ¹ = 2	2	0.301
40	2	2 ² = 4	4	0.602
60	3	2 ³ = 8	8	0.903
80	4	2 ⁴ = 16	16	1.204
100	5	2 ⁵ = 32	32	1.505
120	6	2 ⁶ = 64	64	1.806

^aThe hypothetical culture begins with one cell having a 20-minute generation time.

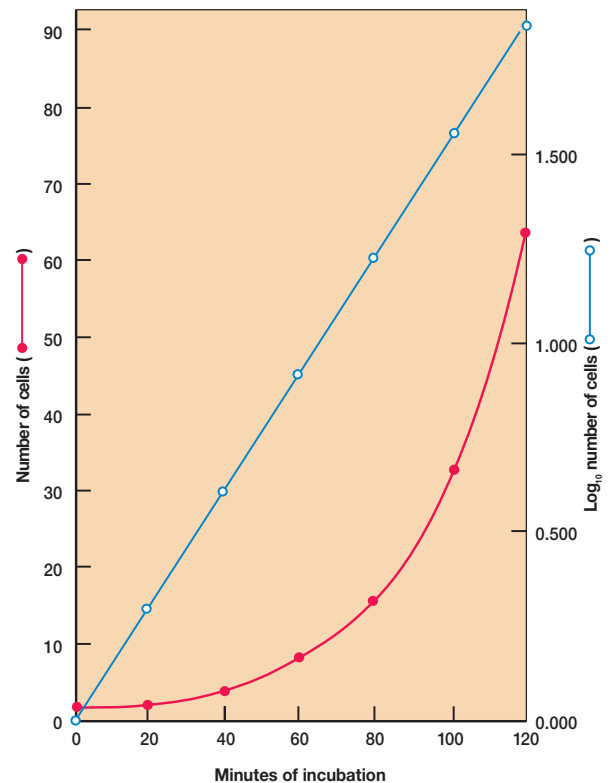


Figure 6.3 Exponential Microbial Growth. The data from table 6.1 for six generations of growth are plotted directly (●—●) and in the logarithmic form (○—○). The growth curve is exponential as shown by the linearity of the log plot.

bling time. This situation can be illustrated with a simple example. Suppose that a culture tube is inoculated with one cell that divides every 20 minutes (table 6.1). The population will be 2 cells after 20 minutes, 4 cells after 40 minutes, and so forth. Because the population is doubling every generation, the increase in population is always 2ⁿ where n is the number of generations. The resulting population increase is exponential or logarithmic (figure 6.3).

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These observations can be expressed as equations for the generation time.

Let N_0 = the initial population number

N_t = the population at time t

n = the number of generations in time t

Then inspection of the results in table 6.1 will show that

$$N_t = N_0 \times 2^n.$$

Solving for n , the number of generations, where all logarithms are to the base 10,

$$\log N_t = \log N_0 + n \cdot \log 2, \text{ and}$$

$$n = \frac{\log N_t - \log N_0}{\log 2} = \frac{\log N_t - \log N_0}{0.301}$$

The rate of growth during the exponential phase in a batch culture can be expressed in terms of the **mean growth rate constant (k)**. This is the number of generations per unit time, often expressed as the generations per hour.

$$k = \frac{n}{t} = \frac{\log N_t - \log N_0}{0.301t}$$

The time it takes a population to double in size—that is, the **mean generation time** or mean doubling time (g), can now be calculated. If the population doubles ($t = g$), then

$$N_t = 2 N_0.$$

Substitute $2N_0$ into the mean growth rate equation and solve for k .

$$k = \frac{\log (2N_0) - \log N_0}{0.301g} = \frac{\log 2 + \log N_0 - \log N_0}{0.301g}$$

$$k = \frac{1}{g}$$

The mean generation time is the reciprocal of the mean growth rate constant.

$$g = \frac{1}{k}$$

The mean generation time (g) can be determined directly from a semilogarithmic plot of the growth data (**figure 6.4**) and the growth rate constant calculated from the g value. The generation time also may be calculated directly from the previous equations. For example, suppose that a bacterial population increases from 10^3 cells to 10^9 cells in 10 hours.

$$k = \frac{\log 10^9 - \log 10^3}{(0.301)(10 \text{ hr})} = \frac{9 - 3}{3.01 \text{ hr}} = 2.0 \text{ generations/hr}$$

$$g = \frac{1}{2.0 \text{ gen./hr}} = 0.5 \text{ hr/gen. or } 30 \text{ min/gen.}$$

Generation times vary markedly with the species of microorganism and environmental conditions. They range from less

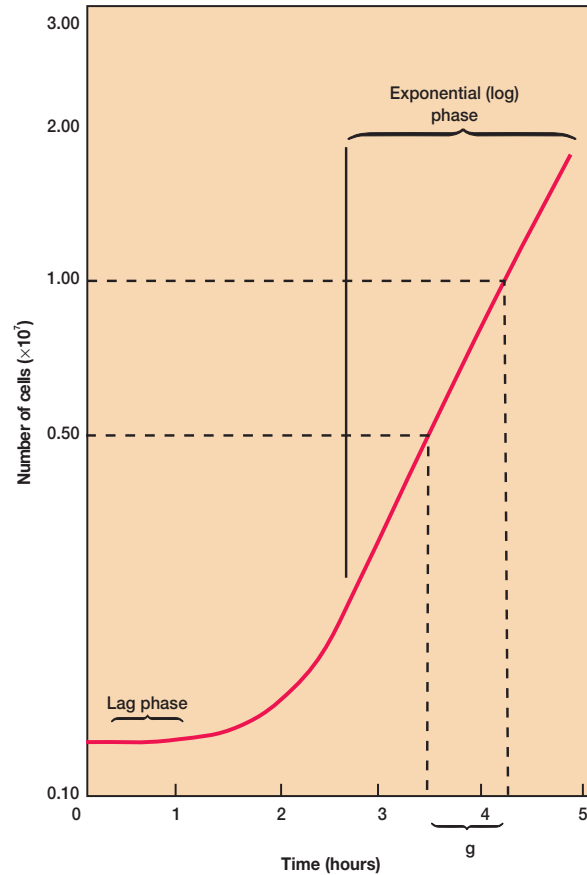


Figure 6.4 Generation Time Determination. The generation time can be determined from a microbial growth curve. The population data are plotted with the logarithmic axis used for the number of cells. The time to double the population number is then read directly from the plot. The log of the population number can also be plotted against time on regular axes.

than 10 minutes (0.17 hours) for a few bacteria to several days with some eucaryotic microorganisms (**table 6.2**). Generation times in nature are usually much longer than in culture.

1. Define growth. Describe the four phases of the growth curve in a closed system and discuss the causes of each.
2. Define balanced growth, unbalanced growth, shift-up experiment, and shift-down experiment.
3. What effect does increasing a limiting nutrient have on the yield of cells and the growth rate?
4. What are the generation or doubling time and the mean growth rate constant? How can they be determined from growth data?

Table 6.2 Generation Times for Selected Microorganisms

Microorganism	Temperature (°C)	Generation Time (Hours)
Bacteria		
<i>Beneckea natriegens</i>	37	0.16
<i>Escherichia coli</i>	40	0.35
<i>Bacillus subtilis</i>	40	0.43
<i>Staphylococcus aureus</i>	37	0.47
<i>Pseudomonas aeruginosa</i>	37	0.58
<i>Clostridium botulinum</i>	37	0.58
<i>Rhodospirillum rubrum</i>	25	4.6–5.3
<i>Anabaena cylindrica</i>	25	10.6
<i>Mycobacterium tuberculosis</i>	37	≈12
<i>Treponema pallidum</i>	37	33
Algae		
<i>Scenedesmus quadricauda</i>	25	5.9
<i>Chlorella pyrenoidosa</i>	25	7.75
<i>Asterionella formosa</i>	20	9.6
<i>Euglena gracilis</i>	25	10.9
<i>Ceratium tripos</i>	20	82.8
Protozoa		
<i>Tetrahymena geleii</i>	24	2.2–4.2
<i>Leishmania donovani</i>	26	10–12
<i>Paramecium caudatum</i>	26	10.4
<i>Acanthamoeba castellanii</i>	30	11–12
<i>Giardia lamblia</i>	37	18
Fungi		
<i>Saccharomyces cerevisiae</i>	30	2
<i>Monilia fructicola</i>	25	30

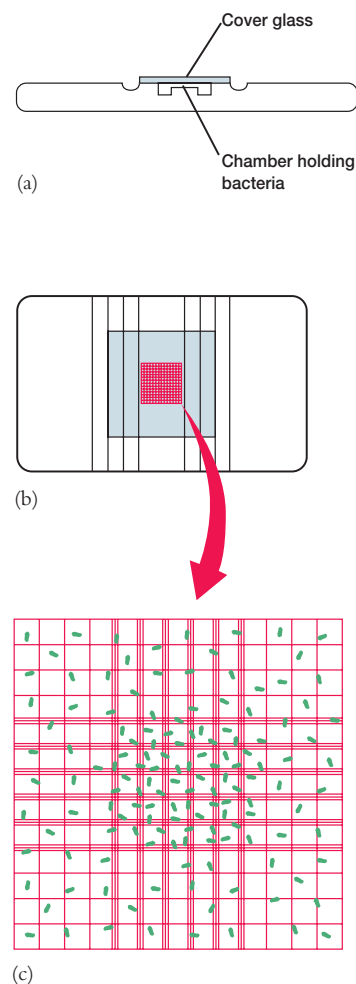


Figure 6.5 The Petroff-Hausser Counting Chamber. (a) Side view of the chamber showing the cover glass and the space beneath it that holds a bacterial suspension. (b) A top view of the chamber. The grid is located in the center of the slide. (c) An enlarged view of the grid. The bacteria in several of the central squares are counted, usually at $\times 400$ to $\times 500$ magnification. The average number of bacteria in these squares is used to calculate the concentration of cells in the original sample. Since there are 25 squares covering an area of 1 mm^2 , the total number of bacteria in 1 mm^2 of the chamber is (number/square)(25 squares). The chamber is 0.02 mm deep and therefore,

$$\text{bacteria/mm}^3 = (\text{bacteria/square})(25 \text{ squares})(50).$$

The number of bacteria per cm^3 is 10^3 times this value. For example, suppose the average count per square is 28 bacteria:

$$\text{bacteria/cm}^3 = (28 \text{ bacteria})(25 \text{ squares})(50)(10^3) = 3.5 \times 10^7.$$

scope is employed. These specially designed slides have chambers of known depth with an etched grid on the chamber bottom (**figure 6.5**). The number of microorganisms in a sample can be calculated by taking into account the chamber's volume and any

6.2 Measurement of Microbial Growth

There are many ways to measure microbial growth to determine growth rates and generation times. Either population mass or number may be followed because growth leads to increases in both. The most commonly employed techniques for growth measurement are examined briefly and the advantages and disadvantages of each noted. No single technique is always best; the most appropriate approach will depend on the experimental situation.

Measurement of Cell Numbers

The most obvious way to determine microbial numbers is through direct counting. Using a counting chamber is easy, inexpensive, and relatively quick; it also gives information about the size and morphology of microorganisms. Petroff-Hausser counting chambers can be used for counting procaryotes; hemocytometers can be used for both procaryotes and eucaryotes. Procaryotes are more easily counted in these chambers if they are stained, or when a phase-contrast or a fluorescence micro-

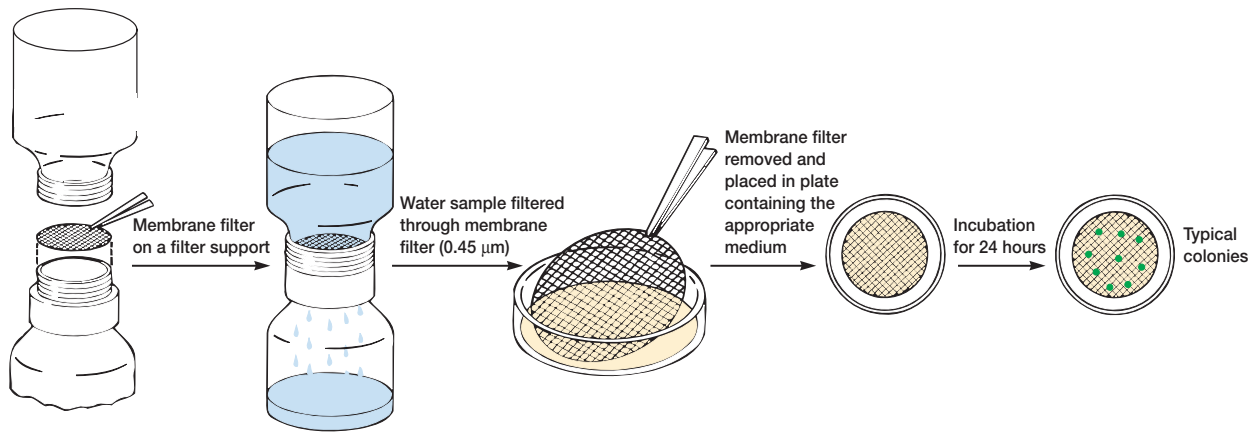


Figure 6.6 The Membrane Filtration Procedure. Membranes with different pore sizes are used to trap different microorganisms. Incubation times for membranes also vary with the medium and microorganism.

sample dilutions required. There are some disadvantages to the technique. The microbial population must be fairly large for accuracy because such a small volume is sampled. It is also difficult to distinguish between living and dead cells in counting chambers without special techniques.

Larger microorganisms such as protozoa, algae, and nonfilamentous yeasts can be directly counted with electronic counters such as the Coulter Counter. The microbial suspension is forced through a small hole or orifice. An electrical current flows through the hole, and electrodes placed on both sides of the orifice measure its electrical resistance. Every time a microbial cell passes through the orifice, electrical resistance increases (or the conductivity drops) and the cell is counted. The Coulter Counter gives accurate results with larger cells and is extensively used in hospital laboratories to count red and white blood cells. It is not as useful in counting bacteria because of interference by small debris particles, the formation of filaments, and other problems.

Counting chambers and electronic counters yield counts of all cells, whether alive or dead. There are also several viable counting techniques, procedures specific for cells able to grow and reproduce. In most viable counting procedures, a diluted sample of bacteria or other microorganisms is dispersed over a solid agar surface. Each microorganism or group of microorganisms develops into a distinct colony. The original number of viable microorganisms in the sample can be calculated from the number of colonies formed and the sample dilution. For example, if 1.0 ml of a 1×10^{-6} dilution yielded 150 colonies, the original sample contained around 1.5×10^8 cells per ml. Usually the count is made more accurate by use of a special colony counter. In this way the spread-plate and pour-plate techniques may be used to find the number of microorganisms in a sample.

Plating techniques are simple, sensitive, and widely used for viable counts of bacteria and other microorganisms in samples of food, water, and soil. Several problems, however, can lead to inac-

curate counts. Low counts will result if clumps of cells are not broken up and the microorganisms well dispersed. Because it is not possible to be absolutely certain that each colony arose from an individual cell, the results are often expressed in terms of **colony forming units (CFU)** rather than the number of microorganisms. The samples should yield between 30 and 300 colonies for best results. Of course the counts will also be low if the agar medium employed cannot support growth of all the viable microorganisms present. The hot agar used in the pour-plate technique may injure or kill sensitive cells; thus spread plates sometimes give higher counts than pour plates. [Spread-plate and pour-plate techniques \(pp. 106–8\)](#)

Microbial numbers are frequently determined from counts of colonies growing on special membrane filters having pores small enough to trap bacteria. In the membrane filter technique, a sample is drawn through a special **membrane filter** ([figure 6.6](#)). The filter is then placed on an agar medium or on a pad soaked with liquid media and incubated until each cell forms a separate colony. A colony count gives the number of microorganisms in the filtered sample, and special media can be used to select for specific microorganisms ([figure 6.7](#)). This technique is especially useful in analyzing aquatic samples. [Analysis of water purity \(pp. 653–57\)](#)

Membrane filters also are used to count bacteria directly. The sample is first filtered through a black polycarbonate membrane filter to provide a good background for observing fluorescent objects. The bacteria then are stained with a fluorescent dye such as acridine orange or DAPI and observed microscopically. Acridine orange-stained microorganisms glow orange or green and are easily counted with an epifluorescence microscope ([see section 2.2](#)). Usually the counts obtained with this approach are much higher than those from culture techniques because some of the bacteria are dead. Commercial kits that use fluorescent reagents to stain live and dead cells differently are now available. This makes it possible to directly count the number of live and dead microorganisms in a sample ([see figure 2.13d](#)).

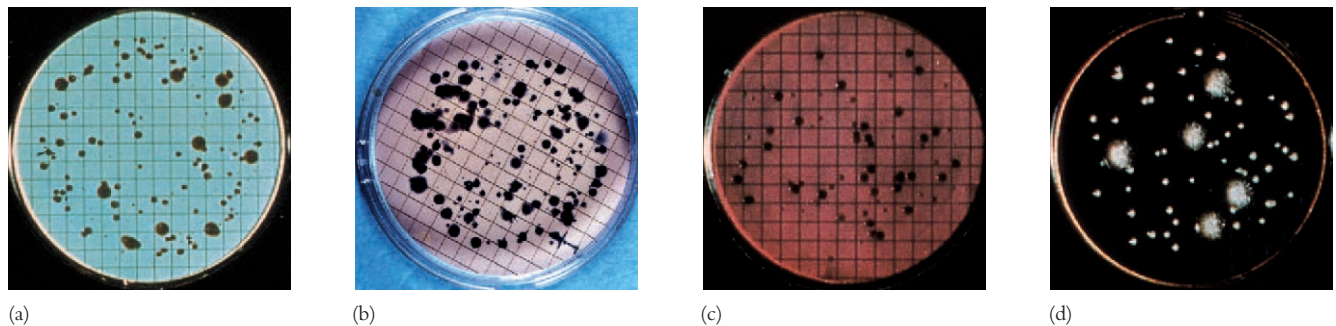


Figure 6.7 Colonies on Membrane Filters. Membrane-filtered samples grown on a variety of media. (a) Standard nutrient media for a total bacterial count. An indicator colors colonies red for easy counting. (b) Fecal coliform medium for detecting fecal coliforms that form blue colonies. (c) m-Endo agar for detecting *E. coli* and other coliforms that produce colonies with a green sheen. (d) Wort agar for the culture of yeasts and molds.

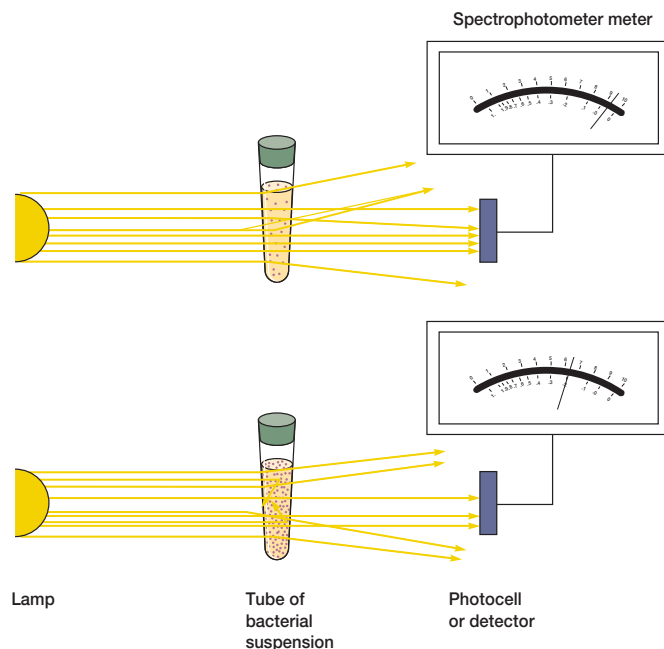


Figure 6.8 Turbidity and Microbial Mass Measurement.

Determination of microbial mass by measurement of light absorption. As the population and turbidity increase, more light is scattered and the absorbance reading given by the spectrophotometer increases. The spectrophotometer meter has two scales. The bottom scale displays absorbance and the top scale, percent transmittance. Absorbance increases as percent transmittance decreases.

Measurement of Cell Mass

Increases in the total cell mass, as well as in cell numbers, accompany population growth. Therefore techniques for measuring changes in cell mass can be used in following growth. The most direct approach is the determination of microbial dry weight. Cells growing in liquid medium are collected by centrifugation, washed, dried in an oven, and weighed. This is an especially useful technique for measuring the growth of fungi. It is time consuming, however, and not very sensitive. Because bacteria weigh so little, it may be necessary to centrifuge several hundred milliliters of culture to collect a sufficient quantity.

More rapid, sensitive techniques depend on the fact that microbial cells scatter light striking them. Because microbial cells

in a population are of roughly constant size, the amount of scattering is directly proportional to the biomass of cells present and indirectly related to cell number. When the concentration of bacteria reaches about 10 million cells (10^7) per ml, the medium appears slightly cloudy or turbid. Further increases in concentration result in greater turbidity and less light is transmitted through the medium. The extent of light scattering can be measured by a spectrophotometer and is almost linearly related to bacterial concentration at low absorbance levels (figure 6.8). Thus population growth can be easily measured spectrophotometrically as long as the population is high enough to give detectable turbidity.

If the amount of a substance in each cell is constant, the total quantity of that cell constituent is directly related to the total

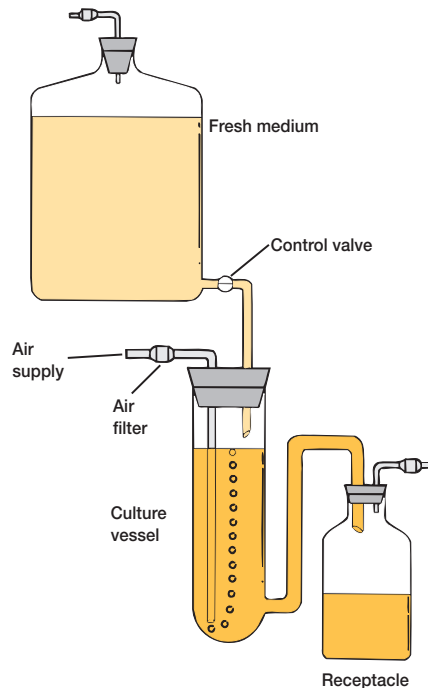


Figure 6.9 A Continuous Culture System: The Chemostat.

Schematic diagram of the system. The fresh medium contains a limiting amount of an essential nutrient. Growth rate is determined by the rate of flow of medium through the culture vessel.

microbial cell mass. For example, a sample of washed cells collected from a known volume of medium can be analyzed for total protein or nitrogen. An increase in the microbial population will be reflected in higher total protein levels. Similarly, chlorophyll determinations can be used to measure algal populations, and the quantity of ATP can be used to estimate the amount of living microbial mass.

1. Briefly describe each technique by which microbial population numbers may be determined and give its advantages and disadvantages.
2. Why are plate count results often expressed as colony forming units?

6.3 The Continuous Culture of Microorganisms

Up to this point the focus has been on closed systems called batch cultures in which nutrient supplies are not renewed nor wastes removed. Exponential growth lasts for only a few generations and soon the stationary phase is reached. However, it is possible to grow microorganisms in an open system, a system with constant environmental conditions maintained through continual provi-

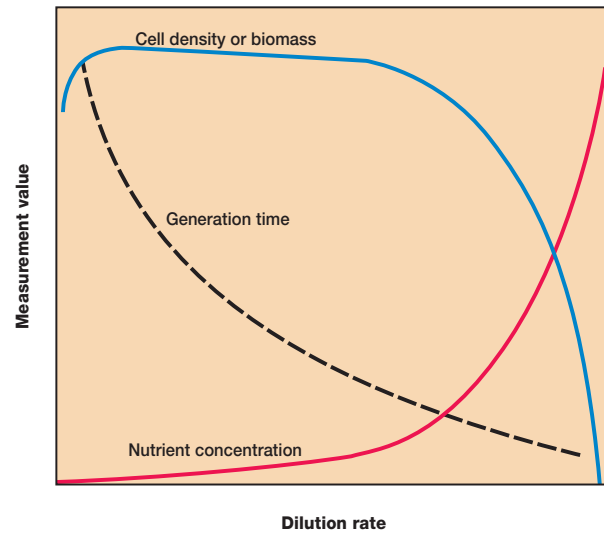


Figure 6.10 Chemostat Dilution Rate and Microbial Growth. The effects of changing the dilution rate in a chemostat.

sion of nutrients and removal of wastes. These conditions are met in the laboratory by a **continuous culture system**. A microbial population can be maintained in the exponential growth phase and at a constant biomass concentration for extended periods in a continuous culture system.

The Chemostat

Two major types of continuous culture systems commonly are used: (1) chemostats and (2) turbidostats. A **chemostat** is constructed so that sterile medium is fed into the culture vessel at the same rate as the media containing microorganisms is removed (**figure 6.9**). The culture medium for a chemostat possesses an essential nutrient (e.g., an amino acid) in limiting quantities. Because of the presence of a limiting nutrient, the growth rate is determined by the rate at which new medium is fed into the growth chamber, and the final cell density depends on the concentration of the limiting nutrient. The rate of nutrient exchange is expressed as the dilution rate (D), the rate at which medium flows through the culture vessel relative to the vessel volume, where f is the flow rate (ml/hr) and V is the vessel volume (ml).

$$D = f/V$$

For example, if f is 30 ml/hr and V is 100 ml, the dilution rate is 0.30 hr^{-1} .

Both the microbial population level and the generation time are related to the dilution rate (**figure 6.10**). The microbial population density remains unchanged over a wide range of dilution rates. The generation time decreases (i.e., the growth rate rises) as the dilution rate increases. The limiting nutrient will be almost

completely depleted under these balanced conditions. If the dilution rate rises too high, the microorganisms can actually be washed out of the culture vessel before reproducing because the dilution rate is greater than the maximum growth rate. The limiting nutrient concentration rises at higher dilution rates because fewer microorganisms are present to use it.

At very low dilution rates, an increase in D causes a rise in both cell density and the growth rate. This is because of the effect of nutrient concentration on the growth rate, sometimes called the Monod relationship (figure 6.2*b*). Only a limited supply of nutrient is available at low dilution rates. Much of the available energy must be used for cell maintenance, not for growth and reproduction. As the dilution rate increases, the amount of nutrients and the resulting cell density rise because energy is available for both maintenance and growth. The growth rate increases when the total available energy exceeds the **maintenance energy**.

The Turbidostat

The second type of continuous culture system, the **turbidostat**, has a photocell that measures the absorbance or turbidity of the culture in the growth vessel. The flow rate of media through the vessel is automatically regulated to maintain a predetermined turbidity or cell density. The turbidostat differs from the chemostat in several ways. The dilution rate in a turbidostat varies rather than remaining constant, and its culture medium lacks a limiting nutrient. The turbidostat operates best at high dilution rates; the chemostat is most stable and effective at lower dilution rates.

Continuous culture systems are very useful because they provide a constant supply of cells in exponential phase and growing at a known rate. They make possible the study of microbial growth at very low nutrient levels, concentrations close to those present in natural environments. These systems are essential for research in many areas—for example, in studies on interactions between microbial species under environmental conditions resembling those in a freshwater lake or pond. Continuous systems also are used in food and industrial microbiology.

1. How does an open system differ from a closed culture system or batch culture?
2. Describe how the two different kinds of continuous culture systems, the chemostat and turbidostat, operate.
3. What is the dilution rate? What is maintenance energy?

6.4 The Influence of Environmental Factors on Growth

As we have seen (pp. 114–15), microorganisms must be able to respond to variations in nutrient levels, and particularly to nutrient limitation. The growth of microorganisms also is greatly affected by the chemical and physical nature of their surroundings. An understanding of environmental influences aids in the control of microbial growth and the study of the ecological distribution of microorganisms.

The ability of some microorganisms to adapt to extreme and inhospitable environments is truly remarkable. Prokaryotes are present anywhere life can exist. Many habitats in which prokaryotes thrive would kill most other organisms. Prokaryotes such as *Bacillus infernus* even seem able to live over 1.5 miles below the Earth's surface, without oxygen and at temperatures above 60°C. Microorganisms that grow in such harsh conditions are often called **extremophiles**.

In this section we shall briefly review how some of the most important environmental factors affect microbial growth. Major emphasis will be given to solutes and water activity, pH, temperature, oxygen level, pressure, and radiation. **Table 6.3** summarizes the way in which microorganisms are categorized in terms of their response to these factors.

Solutes and Water Activity

Because a selectively permeable plasma membrane separates microorganisms from their environment, they can be affected by changes in the osmotic concentration of their surroundings. If a microorganism is placed in a hypotonic solution (one with a lower osmotic concentration), water will enter the cell and cause it to burst unless something is done to prevent the influx. The osmotic concentration of the cytoplasm can be reduced by use of inclusion bodies (see pp. 49–52). Prokaryotes also can contain pressure-sensitive channels that open to allow solute escape when the osmolarity of the environment becomes much lower than that of the cytoplasm.

Most bacteria, algae, and fungi have rigid cell walls that maintain the shape and integrity of the cell. When microorganisms with rigid cell walls are placed in a hypertonic environment, water leaves and the plasma membrane shrinks away from the wall, a process known as plasmolysis. This dehydrates the cell and may damage the plasma membrane; the cell usually becomes metabolically inactive and ceases to grow.

Many microorganisms keep the osmotic concentration of their protoplasm somewhat above that of the habitat by the use of compatible solutes, so that the plasma membrane is always pressed firmly against their cell wall. **Compatible solutes** are solutes that are compatible with metabolism and growth when at high intracellular concentrations. Most prokaryotes increase their internal osmotic concentration in a hypertonic environment through the synthesis or uptake of choline, betaine, proline, glutamic acid, and other amino acids; elevated levels of potassium ions are also involved to some extent. Algae and fungi employ sucrose and polyols—for example, arabitol, glycerol, and mannitol—for the same purpose. Polyols and amino acids are ideal solutes for this function because they normally do not disrupt enzyme structure and function. A few prokaryotes like *Halobacterium salinarum* raise their osmotic concentration with potassium ions (sodium ions are also elevated but not as much as potassium). *Halobacterium*'s enzymes have been altered so that they actually require high salt concentrations for normal activity (see section 20.3). Since protozoa do not have a cell wall, they must use contractile vacuoles (see figure 27.3) to eliminate excess water when living in hypotonic environments. **Osmosis and the protective function of the cell wall** (p. 61)

Table 6.3 Microbial Responses to Environmental Factors

Descriptive Term	Definition	Representative Microorganisms
Solute and Water Activity		
Osmotolerant	Able to grow over wide ranges of water activity or osmotic concentration	<i>Staphylococcus aureus</i> , <i>Saccharomyces rouxii</i>
Halophile	Requires high levels of sodium chloride, usually above about 0.2 M, to grow	<i>Halobacterium</i> , <i>Dunaliella</i> , <i>Ectothiorhodospira</i>
pH		
Acidophile	Growth optimum between pH 0 and 5.5	<i>Sulfolobus</i> , <i>Picrophilus</i> , <i>Ferroplasma</i> , <i>Acontium</i> , <i>Cyanidium caldarium</i>
Neutrophile	Growth optimum between pH 5.5 and 8.0	<i>Escherichia</i> , <i>Euglena</i> , <i>Paramecium</i>
Alkalophile	Growth optimum between pH 8.5 and 11.5	<i>Bacillus alcalophilus</i> , <i>Nastronobacterium</i>
Temperature		
Psychrophile	Grows well at 0°C and has an optimum growth temperature of 15°C or lower	<i>Bacillus psychrophilus</i> , <i>Chlamydomonas nivalis</i>
Psychrotroph	Can grow at 0–7°C; has an optimum between 20 and 30°C and a maximum around 35°C	<i>Listeria monocytogenes</i> , <i>Pseudomonas fluorescens</i>
Mesophile	Has growth optimum around 20–45°C	<i>Escherichia coli</i> , <i>Neisseria gonorrhoeae</i> , <i>Trichomonas vaginalis</i>
Thermophile	Can grow at 55°C or higher; optimum often between 55 and 65°C	<i>Bacillus stearothermophilus</i> , <i>Thermus aquaticus</i> , <i>Cyanidium caldarium</i> , <i>Chaetomium thermophile</i>
Hyperthermophile	Has an optimum between 80 and about 113°C	<i>Sulfolobus</i> , <i>Pyrococcus</i> , <i>Pyrodicticum</i>
Oxygen Concentration		
Obligate aerobe	Completely dependent on atmospheric O ₂ for growth.	<i>Micrococcus luteus</i> , <i>Pseudomonas</i> , <i>Mycobacterium</i> ; most algae, fungi, and protozoa
Facultative anaerobe	Does not require O ₂ for growth, but grows better in its presence.	<i>Escherichia</i> , <i>Enterococcus</i> , <i>Saccharomyces cerevisiae</i>
Aerotolerant anaerobe	Grows equally well in presence or absence of O ₂	<i>Streptococcus pyogenes</i>
Obligate anaerobe	Does not tolerate O ₂ and dies in its presence.	<i>Clostridium</i> , <i>Bacteroides</i> , <i>Methanobacterium</i> , <i>Trepomonas agilis</i>
Microaerophile	Requires O ₂ levels below 2–10% for growth and is damaged by atmospheric O ₂ (20%).	<i>Campylobacter</i> , <i>Spirillum volutans</i> , <i>Treponema pallidum</i>
Pressure		
Barophilic	Growth more rapid at high hydrostatic pressures.	<i>Photobacterium profundum</i> , <i>Shewanella benthica</i> , <i>Methanococcus jannaschii</i>

The amount of water available to microorganisms can be reduced by interaction with solute molecules (the osmotic effect) or by adsorption to the surfaces of solids (the matric effect). Because the osmotic concentration of a habitat has such profound effects on microorganisms, it is useful to be able to express quantitatively the degree of water availability. Microbiologists generally use **water activity** (a_w) for this purpose (water availability also may be expressed as water potential, which is related to a_w). The water activity of a solution is 1/100 the relative humidity of the solution (when expressed as a percent). It is also equivalent to the ratio of the solution's vapor pressure (P_{soln}) to that of pure water (P_{water}).

$$a_w = \frac{P_{\text{soln}}}{P_{\text{water}}}$$

The water activity of a solution or solid can be determined by sealing it in a chamber and measuring the relative humidity after the system has come to equilibrium. Suppose after a sample is treated in this way, the air above it is 95% saturated—that is, the air contains 95% of the moisture it would have when equilibrated at the same temperature with a sample of pure water. The relative humidity would be 95% and the sample's water activity, 0.95.

Water activity is inversely related to osmotic pressure; if a solution has high osmotic pressure, its a_w is low.

Microorganisms differ greatly in their ability to adapt to habitats with low water activity (**table 6.4**). A microorganism must expend extra effort to grow in a habitat with a low a_w value because it must maintain a high internal solute concentration to retain water. Some microorganisms can do this and are **osmotolerant**; they will grow over wide ranges of water activity or osmotic concentration. For example, *Staphylococcus aureus* can be cultured in media containing any sodium chloride concentration up to about 3 M. It is well adapted for growth on the skin. The yeast *Saccharomyces rouxii* will grow in sugar solutions with a_w values as low as 0.6. The alga *Dunaliella viridis* tolerates sodium chloride concentrations from 1.7 M to a saturated solution.

Although a few microorganisms are truly osmotolerant, most only grow well at water activities around 0.98 (the approximate a_w for seawater) or higher. This is why drying food or adding large quantities of salt and sugar is so effective in preventing food spoilage. As table 6.4 shows, many fungi are osmotolerant and thus particularly important in the spoilage of salted or dried foods. [Food spoilage \(pp. 966–69\)](#)